



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G1

System Integration and Architecture

System Design Document (SDD)

Project Title: RateFlow

Prepared By: Cruz, John Harley Abellana

Version: 1.1

Date: 3/7/2026

Status: Draft

REVISION HISTORY TABLE

Version	Date	Author	Changes Made	Status
0.1	2/21/2026	John Harley A. Cruz	Initial draft	Draft
1.0	2/28/2026	John Harley A. Cruz	API Contracts / Communication → API Specifications & Interaction Database Design → Database Structure Planning UI/UX → User Interface & User Experience Design Plan → Project Strategy	Done
1.1	3/7/2026	John Harley A. Cruz	Updated History Table	Final

TABLE OF CONTENTS

Contents

EXECUTIVE SUMMARY	4
1.0 INTRODUCTION	5
2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION	5
3.0 NON-FUNCTIONAL REQUIREMENTS	8
4.0 SYSTEM ARCHITECTURE	9
5.0 API CONTRACT & COMMUNICATION	10
Authentication Endpoints	11
User Registration	11
User Login	11
6.0 DATABASE DESIGN	13
7.0 UI/UX DESIGN	14
8.0 PLAN	17

EXECUTIVE SUMMARY

1.1 Project Overview

RateFlow MVP is a simplified service-based rating and feedback management platform that allows users to submit ratings and reviews for services while enabling administrators to monitor and manage feedback efficiently.

1.2 Objectives

1. Develop a fully functional service rating MVP with authentication and feedback submission
2. Implement a **three-tier architecture** using Spring Boot (backend), React (web), and Android (mobile)
3. Create RESTful APIs for communication between frontend and backend systems
4. Allow users to submit, edit, and view ratings and feedback
5. Enable administrators to monitor and manage submitted feedback
6. Design a responsive and user-friendly interface for both web and mobile platforms
7. Deploy system components in a production-ready environment

1.3 Scope

Included Features:

- User registration and authentication (email/password)

- User profile management
- Submit ratings (e.g., 1–5 stars) with written feedback
- View submitted ratings and feedback history
- Admin dashboard for monitoring and managing feedbacks
- Search and filter feedback entries
- RESTful API integration
- Relational database (MySQL)
- Responsive React web interface
- Native Android mobile application
- Forgot Password System
- Google Auth sign in feature

Excluded Features:

- Real-time push notifications
- Advanced analytics dashboard with AI insights
- Payment system integration
- Automated moderation using AI
- Multi-language support
- Third-party API integrations

1.0 INTRODUCTION

1.1 Purpose

This document serves as the comprehensive design specification for the **RateFlow MVP** system. It provides detailed functional and non-functional requirements, system architecture, API specifications, database design, and implementation guidelines.

The purpose of this document is to:

- Clearly define the overall system structure and component interactions

- Establish the architectural framework (frontend, backend, and database layers)
- Specify RESTful API contracts for seamless communication between web, mobile, and backend services
- Outline the database schema and data management strategy
- Provide a development roadmap to guide implementation and deployment

This document ensures that all system components are aligned, integrated properly, and developed according to a consistent technical standard, minimizing integration issues and improving overall system reliability.

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

Project Name: RateFlow MVP

Domain: Service Rating & Feedback Management

Primary Users:

- Customers / End-Users
- Administrators

Problem Statement:

Organizations and service providers need a simple and efficient way to collect, manage, and analyze customer feedback without relying on complex or expensive enterprise feedback systems.

Solution:

A minimalist rating and feedback platform that allows users to submit service ratings and written reviews, while administrators can monitor, manage, and analyze feedback across web and mobile platforms with a consistent user experience.

2.2 Core User Journeys

Journey 1: First-time User Submits Feedback

1. User visits web or mobile application

2. Clicks "**Sign Up**" and creates an account
 3. Logs in successfully
 4. Selects a service to rate
 5. Submits star rating (1–5 stars)
 6. Writes optional feedback comment
 7. Submits feedback
 8. Receives confirmation message
-

Journey 2: Returning User Submits and Views Feedback

1. User logs in with existing credentials
 2. Navigates to service list
 3. Searches for a specific service
 4. Submits a new rating and comment
 5. Views previously submitted feedback
 6. Edits or deletes own feedback
-

Journey 3: Administrator Service and Feedback Management (Ownership-Based)

1. Logs in using admin credentials.
2. Navigates to the **Admin Dashboard**.
3. Open the **Create Event Screen**.
4. **Creates a new service** to be rated by end users.

5. **Views all services they have created.**
 6. **Updates/edit details** of their existing services
 7. **Deletes services** they previously created.
 8. Selects a specific service from the Services Screen.
 9. Navigates to the **Feedback List Screen** for the selected service.
 10. **Views feedback and ratings submitted by end users** for that service.
 11. **Filters feedback** by date or rating.
 12. **Deletes inappropriate or irrelevant feedback** from end users.
-

2.3 Feature List (MoSCoW)

MUST HAVE

- User authentication (register, login, logout)
 - Submit rating (1–5 stars)
 - Submit written feedback
 - Date Created of services
 - View feedback history
 - Admin panel
 - Role-based access control
 - Forgot Password
 - Email Notification System
 - User profile section
 - Admin profile section
 - Google Authentication Sign in
-

SHOULD HAVE

- Search and filter services
- Edit/delete feedback
- Basic rating statistics (average rating per service)

- Responsive design for web and mobile
-

COULD HAVE

- Service categories
 - Basic analytics dashboard
 - Feedback sorting (highest/lowest rating)
 - Notifications
-

WON'T HAVE

- Social login integration
 - AI-based sentiment analysis
 - Multi-language support
 - Third-party API integrations
-

2.4 Detailed Feature Specifications

Feature: User Authentication

Screens: Registration, Login

Fields: Username, Email, Password, Confirm Password

Validation:

- Valid email format
- Password strength requirement
- Unique email check
- Google account must already been registered in the system

API Endpoints:

- `POST api/auth/register`
- `POST api/auth/login`
- `POST api/auth/logout`

- `POST api/auth/google`

Security:

- JWT tokens
 - Password hashing using bcrypt
 - Role-based authorization (User/Admin)
-

Feature: Admin Authentication

Screens:

- Admin Login

Fields:

- Username (fixed: admin)
- Admin Email (fixed by owner gmail)
- Password
- Google account must already been registered in the system

Validation:

- Username must match the predefined admin username ("admin").
- Email must match the predefined secure admin email.
- Password must match the predefined secure admin password.
- Display error message for invalid credentials.

API Endpoints:

`POST api/auth/admin/login`
`POST api/auth/logout`
`POST api/auth/google`

Feature: Service Rating & Feedback

Screens: Service Listing, Service Detail, Submit Rating, Delete own rating

Display:

- Service name
- Description
- Average rating
- Star rating component

Functions:

- Submit rating (1–5 stars)
- Add written feedback
- Edit/delete own feedback

API Endpoints:

- GET `api/services`
 - GET `api/services/{serviceid}`
 - PUT `ratings/{id}`
 - DELETE `ratings/{id}`
 - POST `api/ratings/submit`
 - GET `api/ratings/service/{serviceId}/stats`
 - GET `api/ratings/service/{serviceId}/feedbacks`
 - GET `api/ratings/user/{userId}`
 - DELETE `api/ratings/delete/{ratingId}`
 - GET `/api/ratings/check/user/{userId}/service/{serviceId}`
-

Feature: Admin Service Management

Screens:

- Admin Dashboard
- Create Service Screen
- Access Control Screen
- Manage Service Screen

Fields (Create / Edit Service):

- Service Name
- Description
- Category

Functions:

- Create a new service to be rated
- View list of services created by the logged-in admin
- Edit service details of created services
- Delete services
- View basic rating statistics for each created service

Validation:

- Service name is required
- Service name must be unique per admin
- Only authenticated admins can create services

API Endpoints:

GET	api/services/{serviceId}
POST	api/services/create
PUT	api/services/update/{serviceId}
DELETE	api/services/delete/{serviceId}
GET	api/ratings/user/{userId}

2.5 Acceptance Criteria

◆ USER ACCEPTANCE CRITERIA

AC-1: Successful User Registration

Given I am a new user

When I enter a valid email and strong password

And confirm the password correctly

And submit the registration form

Then my account should be created

And I should be able to log in successfully

AC-2: Successful Rating Submission

Given I am logged in as a user

When I select a service

And submit a rating (1–5 stars)

And optionally write feedback

And click submit

Then the rating should be saved

And it should appear in the service's feedback list

And the service's average rating should update

AC-3: User Can delete their own Feedback/rating

Given I am logged in as a user

Then I navigate to My Ratings page

And it should display the services I rated

When I attempt to delete a rating I submitted

Then the system should allow the user to delete its own rating

Then the system should showcase that the I have deleted my own rating in that specific service

◆ ADMIN ACCEPTANCE CRITERIA

AC-4: Admin Creates and Manages Own Service

Given I am logged in as an administrator

Then I am navigate to the admin dashboard

When I press the create service button from the drawer

Then it would navigate me to the create service page

And when I fill all the required fields

Then press create service button

Then the system will showcase a success message that the service has been created

And the service will be displayed at the admin dashboard page and manage service page

Then I will be navigated back to the admin dashboard page (admin services page)

AC-5: Admin Edits a Service

Given I am logged in as the administrator

And I navigate to the admin dashboard

Then I click the Manage Services button

And I will be navigated to the Manage Service page

And it should display all the created services in that page

When I click edit button in a specific service

Then it should display a modal the edit service modal

When I make changes to the service

And I will click save changes button

Then the changes should be saved successfully

And the updated information should be displayed

AC-6: Admin Deletes a Service

Given I am logged in as the administrator

And I navigate to the admin dashboard

Then I click the Manage Services button

And I will be navigated to the Manage Service page

And it should display all the created services in that page

When I click the delete button on a specific service

Then it should prompt a confirmation for deletion

When I confirm the deletion

Then the service should be permanently removed

And it should no longer appear in the Services Management Screen

AC-7: Only Authenticated Admin Can Grant Admin Access

Given I am logged in as the administrator
And I navigate to the admin dashboard
Then I click the Access Control button
And I will be navigated to the Access Control page
And it should display all registered users
When I select a specific user
And I click the Grant Admin Access button
Then it should prompt a confirmation for granting admin access
When I confirm the action
Then the user's role should be updated to "Admin"
And the user should be able to access admin features

3.0 NON-FUNCTIONAL REQUIREMENTS

3.1 Performance Requirements

- API response time: ≤ 2 seconds for 95% of requests
- Web page load time: ≤ 3 seconds on broadband
- Mobile app cold start: ≤ 3 seconds
- Support 100 concurrent users
- Database queries complete within 500ms

3.2 Security Requirements

- HTTPS for all communications
- JWT token authentication
- Password hashing with bcrypt (salt rounds = 12)
- Admin endpoints require role verification

3.3 Compatibility Requirements

- **Web Browsers:** Chrome, Firefox, Safari, Edge (latest 2 versions)
- **Android:** API Level 24+ (Android 7.0+)
- **Screen Sizes:** Mobile (360px+), Tablet (768px+), Desktop (1024px+)
- **Operating Systems:** Windows 10+, macOS 10.15+, Linux Ubuntu 20.04+

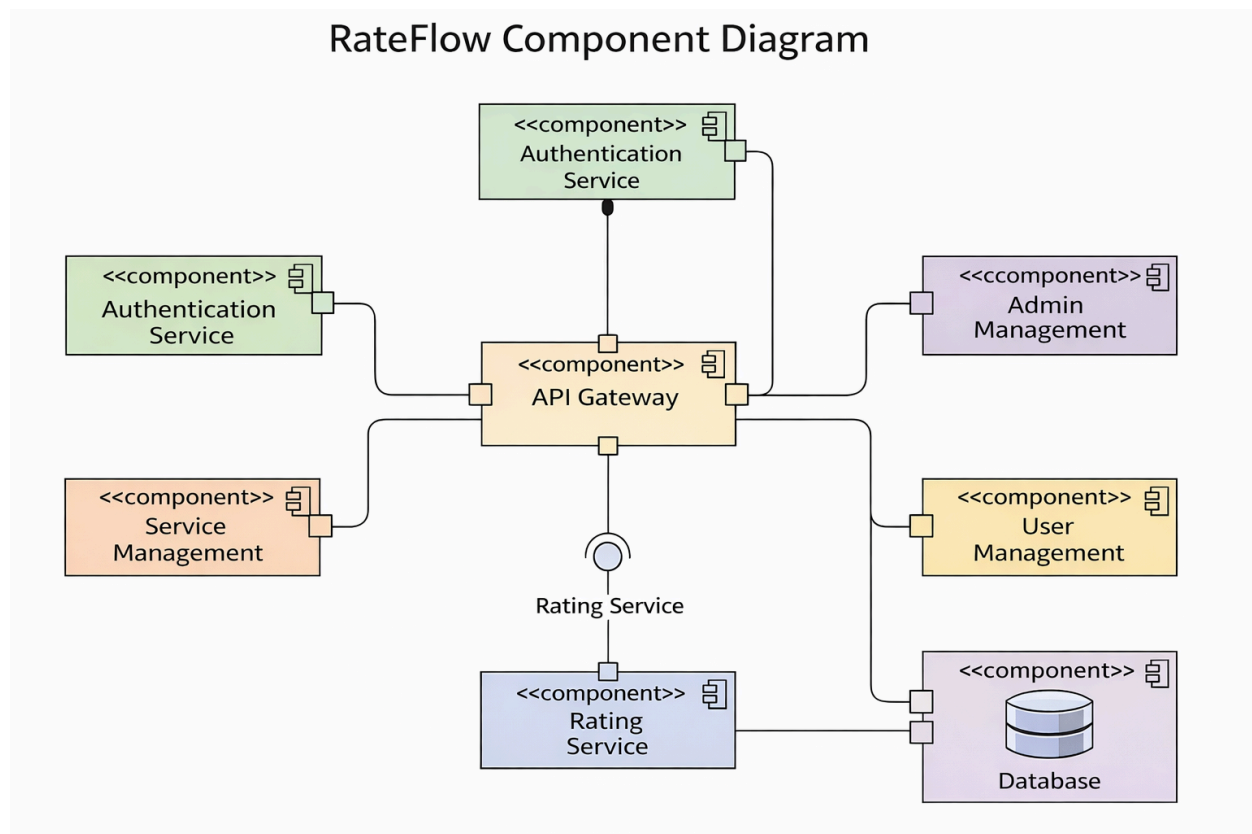
3.4 Usability Requirements

- A user shall be able to submit feedback within 5 minutes
- Admin shall be able to create a new service within 5 minutes
- Interface shall have consistent navigation across all pages
- Clear validation and error messages shall be displayed
- Forms shall prevent invalid input before submission
- Touch targets shall be at least 44x44px on mobile
- The system shall provide simple and intuitive service management for the admin

4.0 SYSTEM ARCHITECTURE

4.1 Component Diagram

Note: This should be a component diagram



Technology Stack:

- **Backend:** Spring Boot DevTools, Spring Web, Spring Data JPA, MySql Driver
- **Database:** PostgreSQL 14+
- **Web Frontend:** React 18
- **Mobile:** Kotlin
- **Build Tools:** Maven (Backend), npm/yarn (Web), Gradle (Android)
- **Deployment:** APK (Mobile) and Render (Web and Backend)
- **Security:** Spring Security, JWT authentication
- **Version Control:** Git/Github

5.0 API CONTRACT & COMMUNICATION

5.1 API Standards

Base URL	http://localhost:8080/api/auth/
Format	JSON for all requests/responses
Authentication	Session-based authentication using HttpSession
Response Structure	{ "success": boolean, "data": object null, "message": string, "timestamp": string }

5.2 Endpoint Specifications

Authentication Endpoints

User Registration

Description	User Registration
API URL	/register
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	{ "username": "<username>", "email": "<email>", "password": "<password>" }
Response Structure	{ "id": 27, }

	<pre> "username": "Testing", "resetToken": null, "resetTokenExpiry": null, "email": "Testing@gmail.com", "password": null, "role": "USER" } </pre>
--	--

User Login

Description	User Login
API URL	/admin/login (handles both user and admin login response)
HTTP Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	<pre> { "email": "<email or username>", "password": "<password>" } </pre>
Response Structure	<pre> { "id": 26, "username": "harleyisgood!", "resetToken": null, "resetTokenExpiry": null, "email": "johnharleycruz592@gmail.com", "password": null, "role": "USER" } </pre>

User Logout

Description	User Logout
API URL	/logout
HTTP Method	POST
Format	JSON for all requests/responses
Authentication	Session-based (User must be logged in)
Request Payload	{}
Response Structure	"Logged out"

Admin Login

Description	Admin Login
API URL	/admin/login (handles both user and admin login response)
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	{ "email": "<email or username>", "password": "<password>" }
Response Structure	{ "id": "<id>", "username": "<username>", "email": "<email>", "role": "<role>" }

Admin Logout

Description	Admin Logout
API URL	/logout
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	Session-based (User must be logged in)
Request Payload	{}
Response Structure	"Logged out"

Get All Services

Description	Retrieve list of all available services
API URL	/services
HTTP Request Method	GET

Format	JSON for all requests/responses
Authentication	None (Public endpoint)
Request Payload	None
Response Structure	<pre>[{ "id": "<id>", "serviceName": "<serviceName>", "serviceCategory": "<serviceCategory>", "serviceDescription": "<serviceDescription>", "createdBy": "<createdBy>", "image": "<byte[]>" }]</pre>

Get Service By ID

Description	Retrieve detailed information of a specific service
API URL	/api/services/{serviceId}
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	None (Public endpoint)
Request Payload	None
Response Structure	<pre>{ "id": "<id>", "serviceName": "<serviceName>", "serviceCategory": "<serviceCategory>", "serviceDescription": "<serviceDescription>", "createdBy": "<createdBy>", "image": "<byte[]>" }], { "error": <null ErrorObject>, "timestamp": <Timestamp> }</pre>

Submit Rating

Description	Submitting a rating to a created service
API URL	/api/ratings/submit
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	Session-based (User must be logged in)
Request Payload	<pre>{ "userId": "<userId>", "serviceId": "<serviceId>", "starRate": "<starRate>", "feedbackText": "<feedbackText>" }</pre>
Response Structure	<pre>{ "success": true, "message": "Rating submitted successfully", "rating": { "ratingId": "<ratingId>", "serviceId": "<serviceId>", "userId": "<userId>", "starRate": "<starRate>", "feedbackText": "<feedbackText>", "dateCreated": "<dateCreated>" } }</pre>

Get Service Feedbacks

Description	Retrieves all ratings and feedbacks for a specific service
API URL	/api/ratings/service/{serviceId}/feedbacks
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	Session-based (User must be logged in)
Request Payload	None

Response Structure	<pre>[{ "ratingId": 1, "userName": "John", "starRate": 5, "feedbackText": "Excellent service!" }]</pre>

Check User Rating

Description	Checks whether a user already rated a service
API URL	/api/ratings/check/user/{userId}/service/{serviceId}
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	Session-based (User must be logged in)
Request Payload	None
Response Structure	<pre>{ "hasRated": true, "rating": 5, "feedback": "Excellent service!" }</pre>

Get Services Rating Statistics

Description	Retrieves rating statistics for a specific service
API URL	/api/ratings/service/{serviceId}/stats
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	Session-based (User must be logged in)
Request Payload	None
Response Structure	<pre>{ "average": 4.5, "total": 20, "distribution": { "1": 1, "2": 0, "3": 2, "4": 5, "5": 12 } }</pre>

Delete Rating

Description	Deleting a rating to a created service
API URL	/delete/{ratingId}
HTTP Request Method	DELETE
Format	JSON for all requests/responses
Authentication	Session-based (User or Admin must be logged in)
Request Payload	None

Response Structure	When user deletes own rating: <pre>{ "success": true, "message": "Your rating has been deleted successfully" }</pre> When admin deletes rating: <pre>{ "success": true, "message": "Feedback deleted successfully by admin" }</pre>
---------------------------	---

Create Service

Description	Create a new service to be rated
API URL	/create
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	Session-based (Admin must be logged in)
Request Payload	<pre>serviceName: <serviceName> serviceCategory: <serviceCategory> serviceDescription: <serviceDescription> (optional) createdBy: <adminUsername> image: <file></pre>
Response Structure	"Service created successfully"

Get All Services (Admin View)

Description	Retrieves all services available in the system for administrative management and review.
API URL	/services
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	Session-based (Admin must be logged in)
Request Payload	None
Response Structure	<pre>[{ "id": "<id>", "serviceName": "<serviceName>", "serviceCategory": "<serviceCategory>", "serviceDescription": "<serviceDescription>", "createdBy": "<createdBy>", "image": "<byte[]>" }]</pre>

Get Service By ID (Admin)

Description	Retrieves detailed information of a specific service for administrative management.
API URL	/serviceId
HTTP Request Method	GET
Format	JSON for all requests/responses

Authentication	Session-based (Admin must be logged in)
Request Payload	None
Response Structure	<pre>{ "id": "<id>", "serviceName": "<serviceName>", "serviceCategory": "<serviceCategory>", "serviceDescription": "<serviceDescription>", "createdBy": "<createdBy>", "image": "<byte[]>" }</pre>

Update Service(Admin)

Description	Allows an authenticated administrator to update the information of a specific service, including its name, category, description, and associated image.
API URL	/update/{serviceId}
HTTP Request Method	PUT
Format	JSON for all requests/responses
Authentication	Session-based (Admin should be logged in)
Request Payload	<pre>serviceName: <serviceName> serviceCategory: <serviceCategory> serviceDescription: <serviceDescription> (optional) createdBy: <adminUsername> image: <file> (optional)</pre>

Response Structure	"Service updated successfully"
---------------------------	--------------------------------

Get user ratings

Description	Retrieves all ratings submitted by a user
API URL	/api/ratings/user/{userId}
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	Session-based (Admin should be logged in)
Request Payload	None
Response Structure	<pre>[{ "ratingId": 1, "serviceId": 5, "starRate": 4, "feedbackText": "Very good service" }]</pre>

Description	Retrieves all ratings submitted by a user

Delete Service (Admin)

Description	Deletes a specific service from the system. This endpoint allows an authenticated administrator to permanently remove a service and its associated data.
API URL	/delete/{serviceId}
HTTP Request Method	DELETE
Format	JSON for all requests/responses
Authentication	Session-based (Admin should be logged in)
Request Payload	None
Response Structure	"Service and its associated ratings deleted successfully"

Google Authentication Login

Description	Google Authentication Login
API URL	/api/auth/google
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	<pre>{ "idToken": "<google_id_token>" }</pre>
Response Structure	<pre>{ "success": true, "role": "USER", "user": { "id": 27, "username": "Testing", "resetToken": null, "resetTokenExpiry": null, "email": "Testing@gmail.com", "password": null, "role": "USER" } }</pre>

Clear User Notification

Description	Deletes all notifications of the currently logged-in user
API URL	/api/user-notifications/clear-all
HTTP Request Method	DELETE
Format	JSON for all requests/responses
Authentication	Session-Based Authentication Required
Request Payload	None
Response Structure	<pre>{ "success": true }</pre>

Delete User Notification

Description	Deletes a specific notification of the logged-in user
API URL	/api/user-notifications/delete/{notificationId}
HTTP Request Method	DELETE

Description	Deletes a specific notification of the logged-in user
Format	JSON for all requests/responses
Authentication	Session-Based Authentication Required
Request Payload	None
Response Structure	<pre>{ "success": true }</pre>

Get User Notification

Description	Retrieves all notifications for the currently logged-in user
API URL	/api/user-notifications
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	Session-Based Authentication Required
Request Payload	None
Response Structure	[

Description	Retrieves all notifications for the currently logged-in user
	<pre>{ "id": 1, "message": "You rated a service with 5 stars", "type": "SERVICE_RATING", "createdAt": "2026-05-07T10:30:00" }</pre>

Admin Notification Management

Description	Retrieves all admin notifications
API URL	/api/notifications
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	Session-Based Authentication Required
Request Payload	None
Response Structure	<pre>[{ "id": 1, "message": "New service created", "createdAt": "2026-05-07T10:30:00" }]</pre>

Description	Retrieves all admin notifications
	]

Delete Notification

Description	Deletes a specific notification by ID
API URL	/api/notifications/delete/{id}
HTTP Request Method	DELETE
Format	JSON for all requests/responses
Authentication	Session-Based Authentication Required
Request Payload	None
Response Structure	<pre>{ "success": true } { "error": "Not authenticated" } { "error": "Access denied" }</pre>

Description	Deletes a specific notification by ID

Clear Notification

Description	Removes all admin notifications
API URL	/api/notifications/delete/{id}
HTTP Request Method	DELETE
Format	JSON for all requests/responses
Authentication	Session-Based Authentication Required
Request Payload	None
Response Structure	<pre> { "success": true } { "error": "Not authenticated" } { "error": "Access denied" } </pre>

5.3 Error Handling

HTTP Status Codes

- 200 OK - Successful request
- 201 Created - Resource created
- 400 Bad Request - Invalid input
- 401 Unauthorized - Authentication required/failed
- 403 Forbidden - Insufficient permissions
- 404 Not Found - Resource doesn't exist
- 409 Conflict - Duplicate resource
- 500 Internal Server Error - Server error

Error Response Format

All API errors follow a consistent response structure:

	<pre>{ "success": false, "data": null, "error": { "code": "string", "message": "string", "details": "string object null" }, "timestamp": "string" }</pre>
--	---

Error Code Examples

Example 1	<pre>json { "success": false, "data": null,</pre>
-----------	---

	<pre> "error": { "code": "AUTH-001", "message": "Invalid credentials", "details": "Email or password is incorrect" }, "timestamp": "2024-01-28T10:30:00Z" } </pre>
Example 2	<pre> { "success": false, "data": null, "error": { "code": "VALID-001", "message": "Validation failed", "details": { "email": "Email is required", "password": "Must be at least 8 characters" } }, "timestamp": "2024-01-28T10:30:00Z" } </pre>

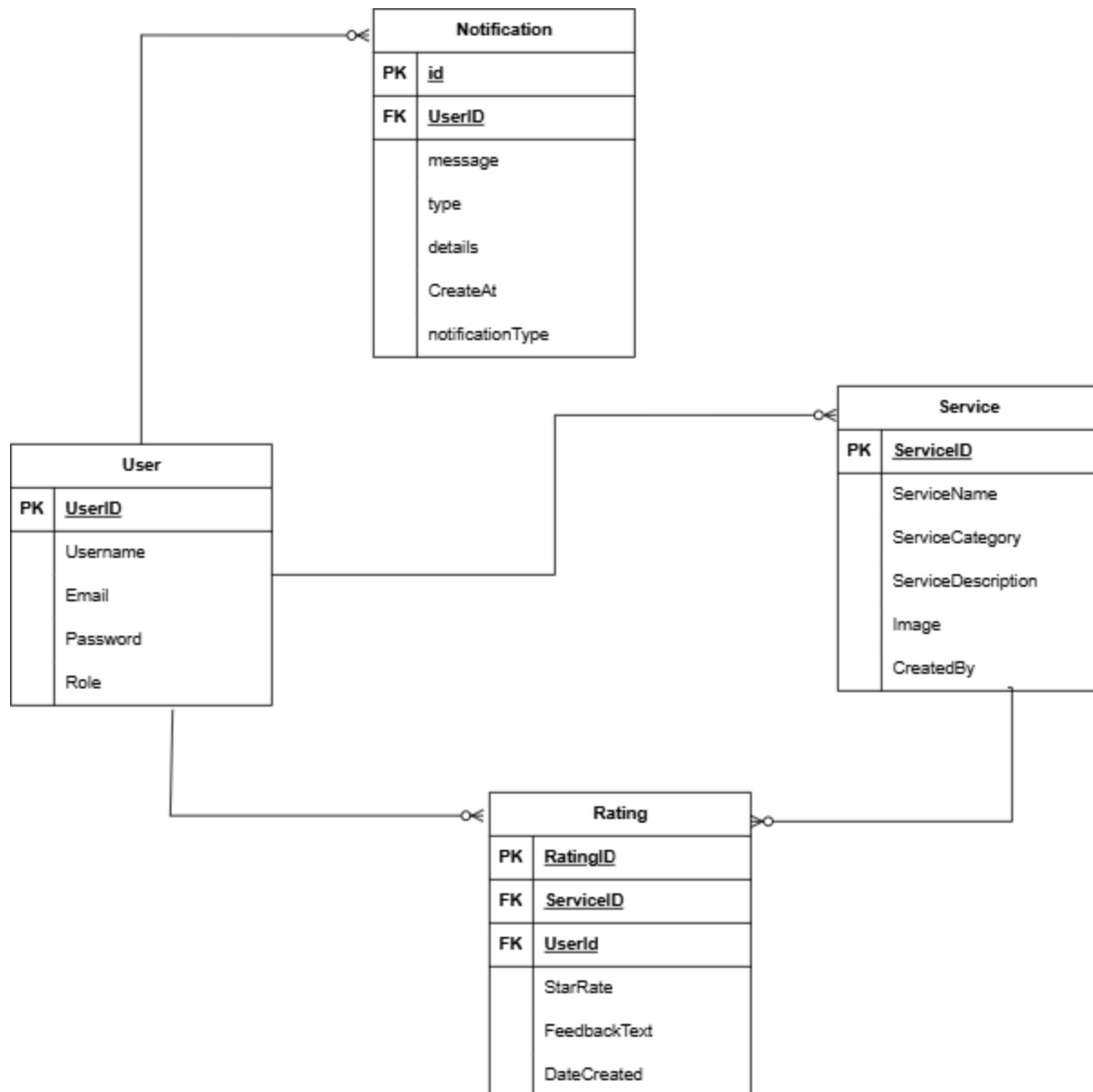
Common Error Codes

- AUTH-001: Invalid credentials
- AUTH-002: Token expired
- AUTH-003: Insufficient permissions
- VALID-001: Validation failed
- DB-001: Resource not found
- DB-002: Duplicate entry
- BUSINESS-001: User already rated this service
- SYSTEM-001: Internal server error

6.0 DATABASE DESIGN

6.1 Entity Relationship Diagram

Note: This should be an ERD (Database Schema)



Detailed Relationships:

One-to-Many: User → Service

One Admin (User with Role = Admin) can create multiple Services.
Each Service is created by exactly one Admin (via **createdBy**).

One-to-Many: User → Rating

One User can submit multiple Ratings.
Each Rating belongs to exactly one User.

One-to-Many: Service → Rating

One Service can receive multiple Ratings.
Each Rating belongs to exactly one Service.

One-to-Many: User → Notification

One User can receive multiple Notifications.
Each Notification belongs to exactly one User.

Many-to-One: Rating → User

Many Ratings can be submitted by one User.

Many-to-One: Rating → Service

Many Ratings can belong to one Service.

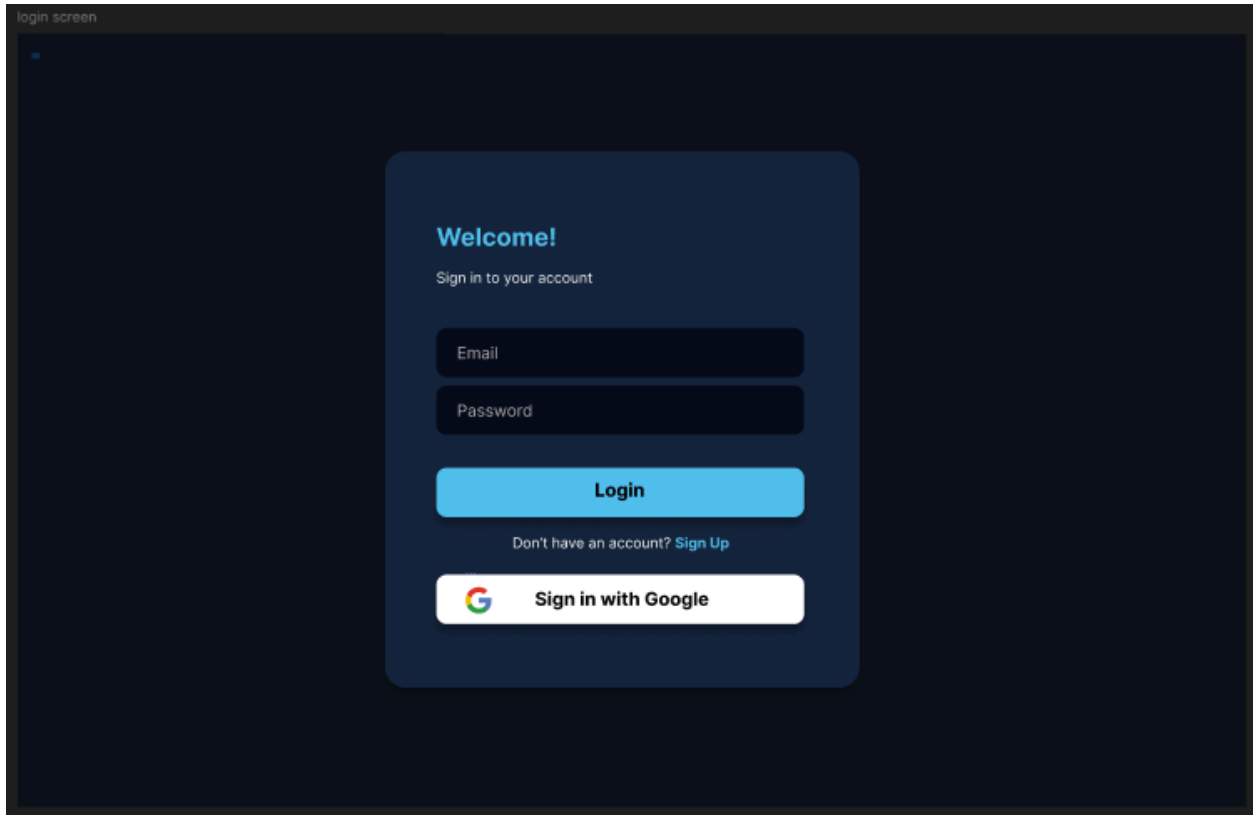
Many-to-One: Notification → User

Many Notifications can belong to one User.

7.0 UI/UX DESIGN

7.1 Web Application Wireframes

User Login Page



Header: Minimal / No Navigation Bar

Content: Authentication Card (Centered on screen)

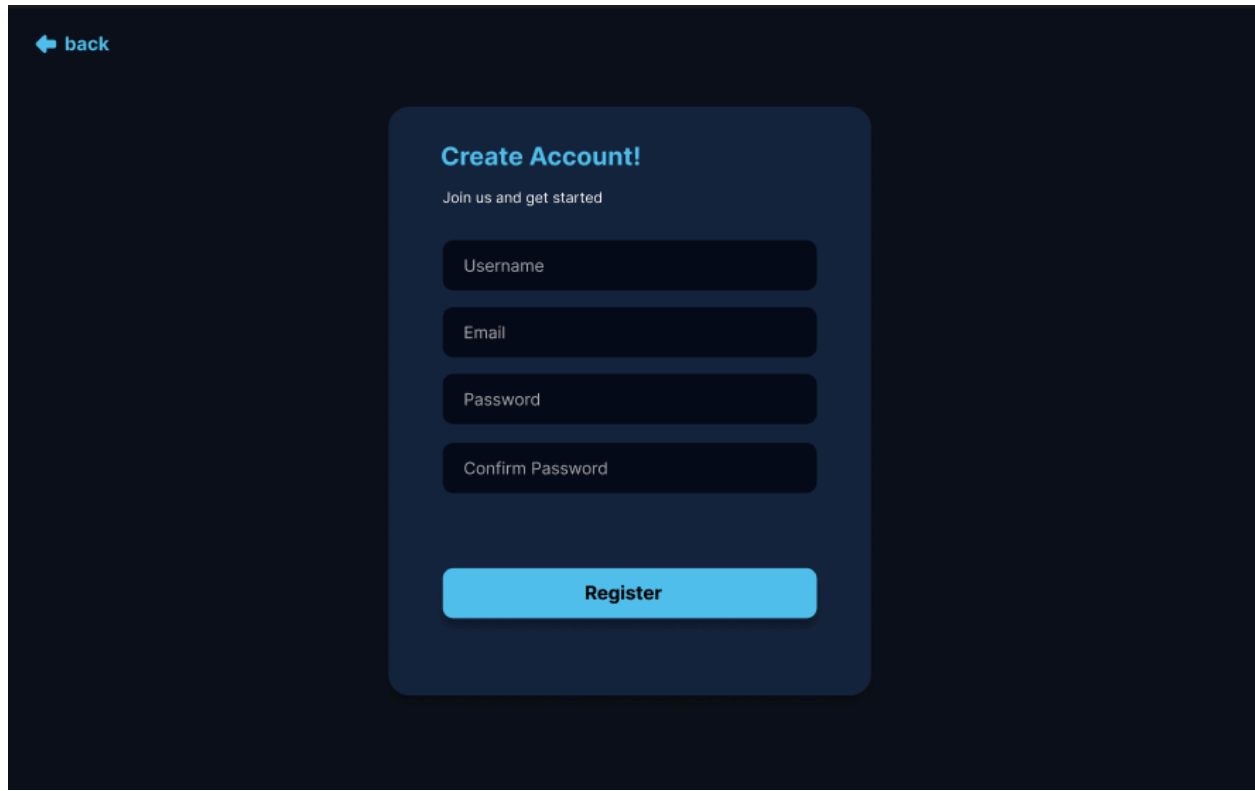
Authentication Card Components:

- Title: "Welcome!"
- Subtitle: "Sign in to your account"
- Email Input Field
- Password Input Field
- "Login" Button (Primary CTA)
- "Sign Up" Link (Secondary Action – below button)
- "Sign In with Google" Button

Background: Dark gradient background (Full screen)

Footer: None (Clean authentication-focused layout)

User Registration Page

A dark-themed user registration page. In the top-left corner, there is a blue back arrow icon followed by the text "back". Centered on the screen is a registration card with a dark blue background and rounded corners. The card contains the title "Create Account!" in light blue, followed by the subtitle "Join us and get started" in a smaller, lighter font. Below the subtitle are four input fields: "Username", "Email", "Password", and "Confirm Password", each with a light blue placeholder text. At the bottom of the card is a prominent blue "Register" button with white text.

Header: Back Button (Top-left corner)

Content (Centered Card Layout): Registration Card (Centered on screen)

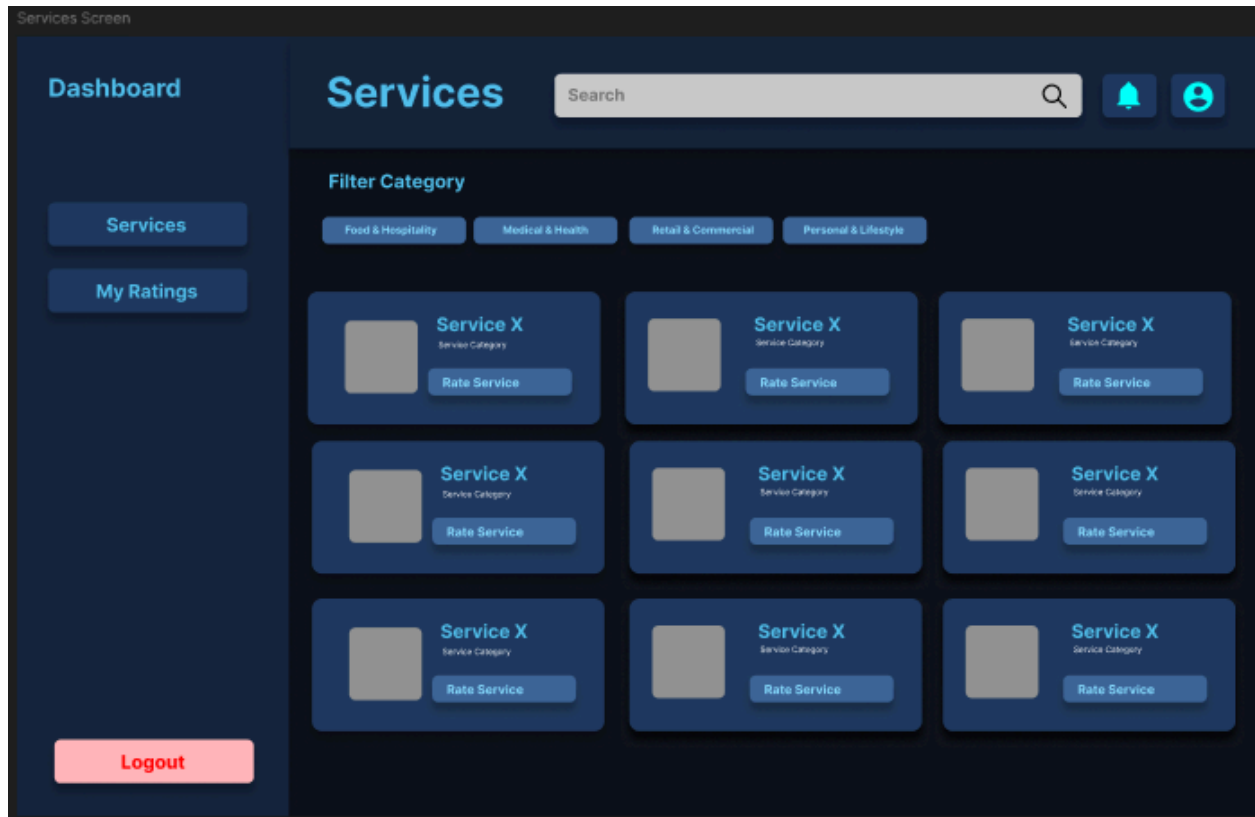
Registration Card Components:

- Title: "Create Account!"
- Subtitle: "Join us and get started"
- Username Input Field
- Email Input Field
- Password Input Field
- Confirm Password Input Field
- "Register" Button (Primary CTA)

Background: Dark gradient background (Full screen)

Footer: None (Authentication-focused layout)

User Dashboard Page



Header:

Page Title ("Services") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title, Services, My Ratings, Logout Button

Content:

Filter Category Buttons (Food & Hospitality, Medical & Health, Retail & Commercial, Personal & Lifestyle)

Service Grid (3 columns desktop)

Each Service Card:

Image / Thumbnail

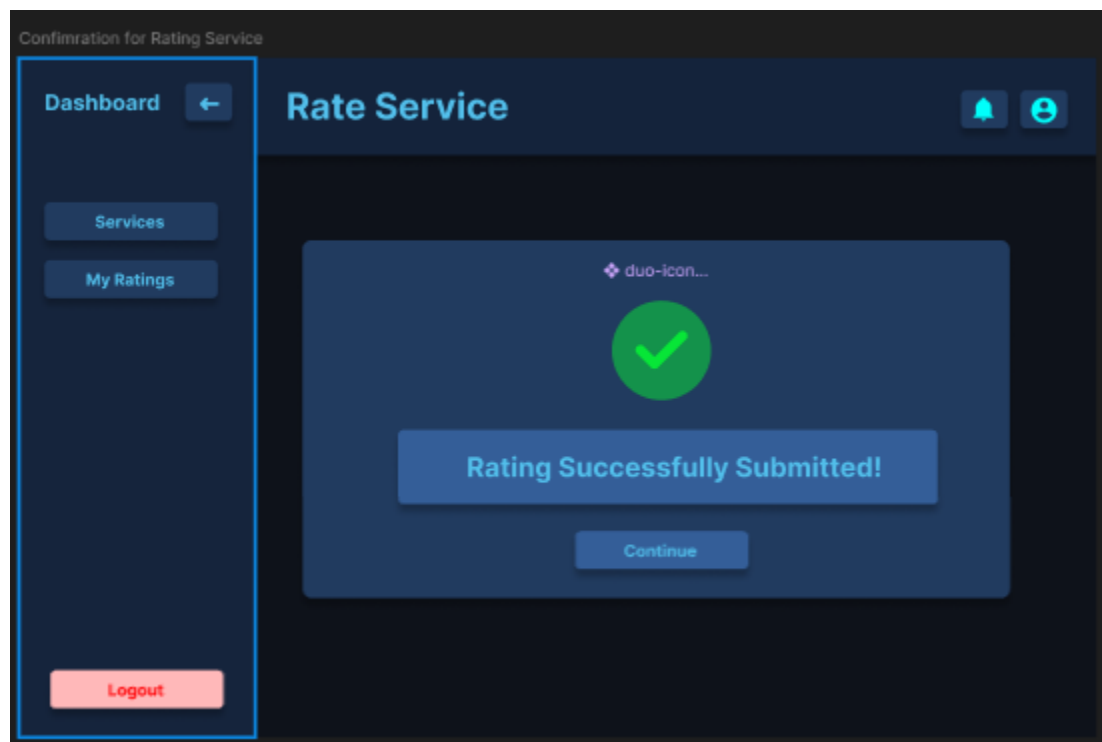
Service Name

Service Category
"Rate Service" Button

Layout:

Left Sidebar + Top Navbar + Main Content Grid
Dark-themed UI

Rating Confirmation Page



Header:

Page Title ("Rate Service") | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title

- Services
- My Ratings
- Logout Button

Content (Centered Card):

Success Icon (Checkmark)

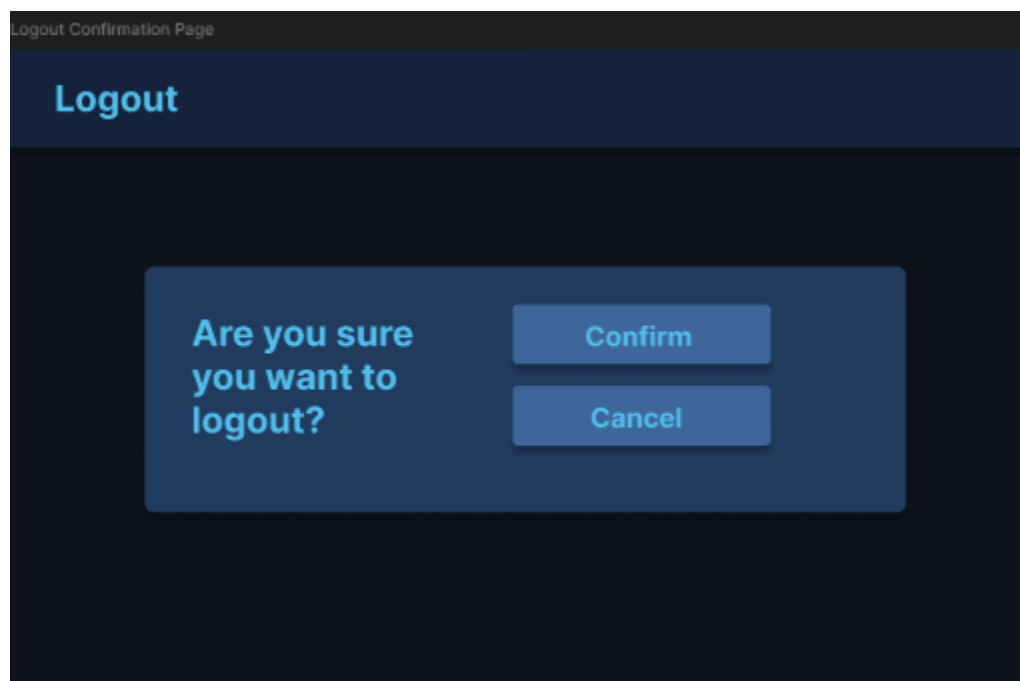
Message: "Rating Successfully Submitted!"

"Continue" Button

Layout:

Left Sidebar + Top Navbar + Centered Confirmation Card

Dark-themed UI

Logout Confirmation Modal**Header:**

Page Title ("Logout")

Content (Centered Card):

Message: "Are you sure you want to logout?"

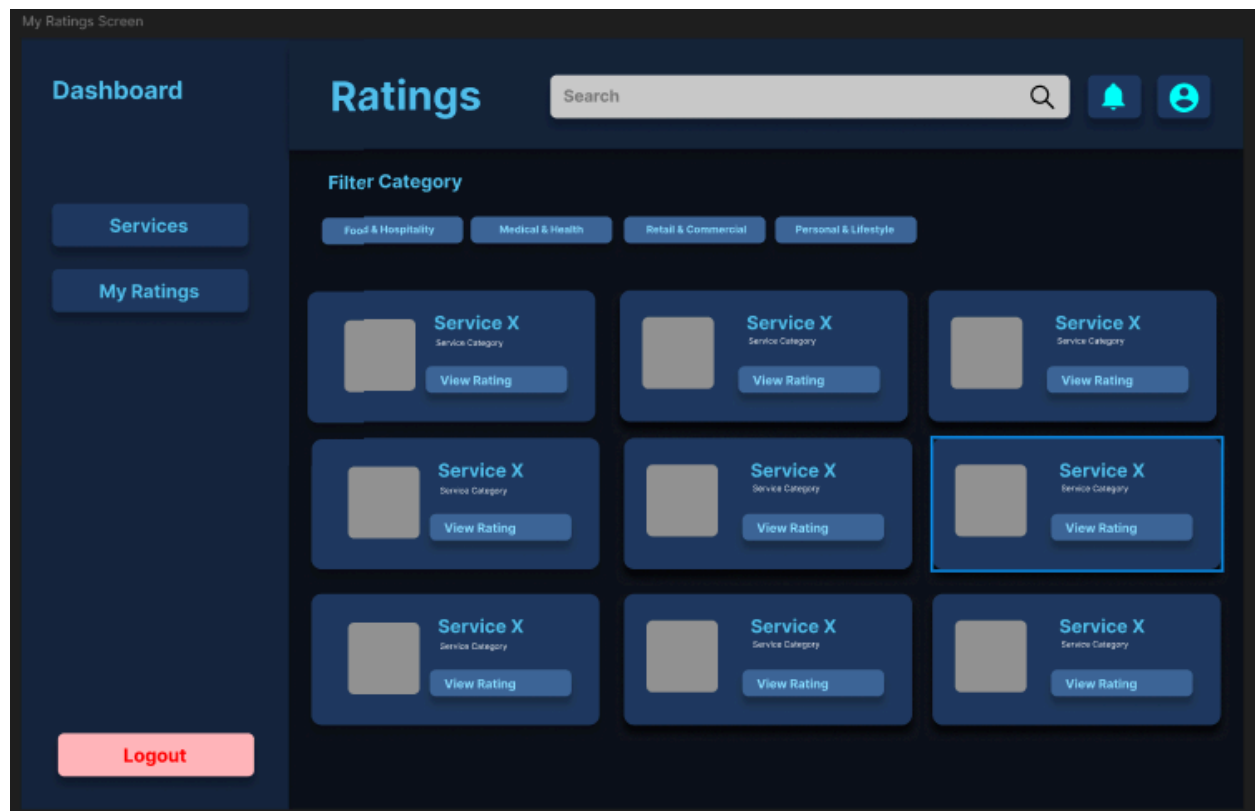
Confirm Button

Cancel Button

Layout:

Top Header + Centered Confirmation Card
Dark-themed UI

My Ratings Page



Header:

Page Title ("Ratings") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title

- Services

- My Ratings (Active)
Logout Button

Content:

Filter Category Buttons (Food & Hospitality, Medical & Health, Retail & Commercial, Personal & Lifestyle)

Ratings Grid (3 columns desktop)

Each Rating Card:

Service Image / Thumbnail

Service Name

Service Category

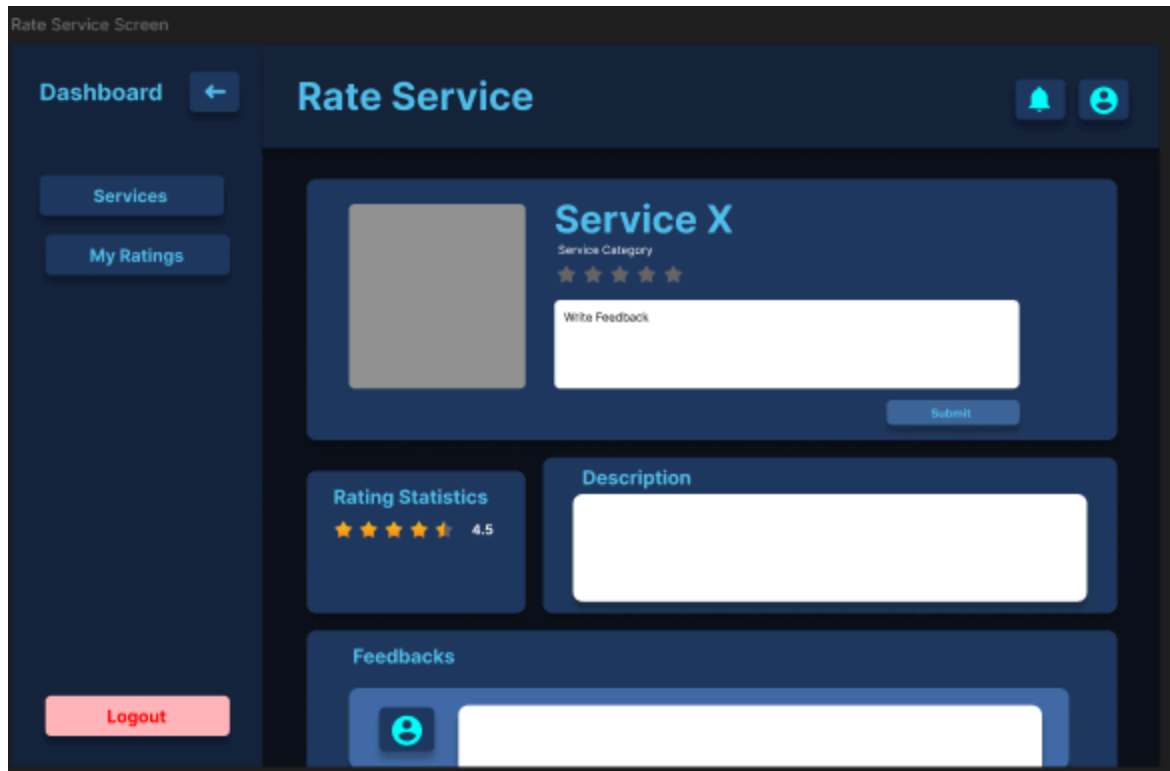
"View Rating" Button

Layout:

Left Sidebar + Top Navbar + Main Content Grid

Dark-themed UI

Submit Rating Page



Header:

Page Title ("Rate Service") | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title

- Services
- My Ratings
- Logout Button

Content:

Service Details Section:

Service Image / Thumbnail
 Service Name
 Service Category
 User Feedback Text
 "Edit Feedback" Button

Rating Statistics Section:

Star Rating Display

Average Rating Value

Description Section:

Service Description Box

Feedbacks Section:

List of User Feedback Cards

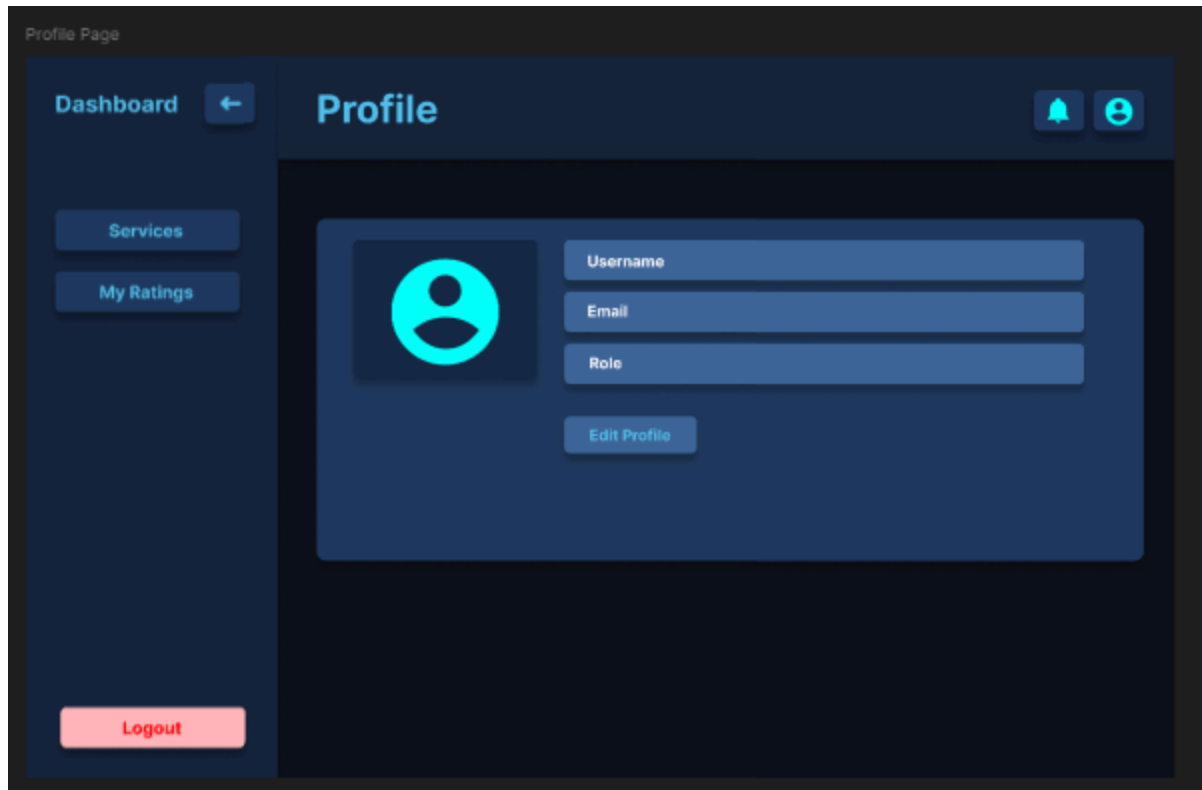
- User Icon
- Star Rating
- Written Feedback

Layout:

Left Sidebar + Top Navbar + Multi-section Content Cards

Dark-themed UI

User Profile Page



Header:

Page Title ("Profile") | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title

- Services
- My Ratings
- Logout Button

Content (Profile Card):

User Avatar / Profile Icon

Username Display

Email Display Field

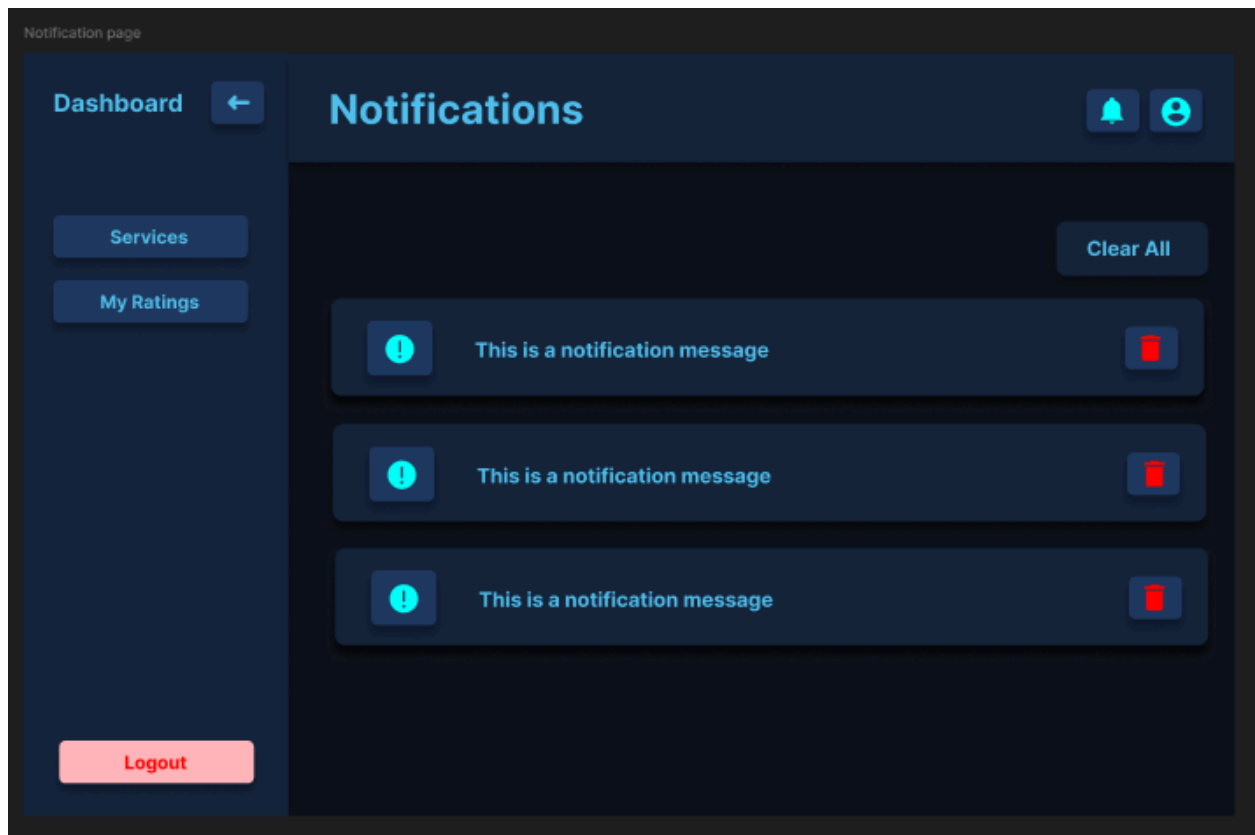
"Edit Profile" Button

Layout:

Left Sidebar + Top Navbar + Centered Profile Card

Dark-themed UI

Notification Page



Header:

Page Title ("Notifications") | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title

- Services
- My Ratings
- Logout Button

Content:

"Clear All" Button

Notification List (Vertical Layout)

Each Notification Card:

Notification Icon

Notification Message

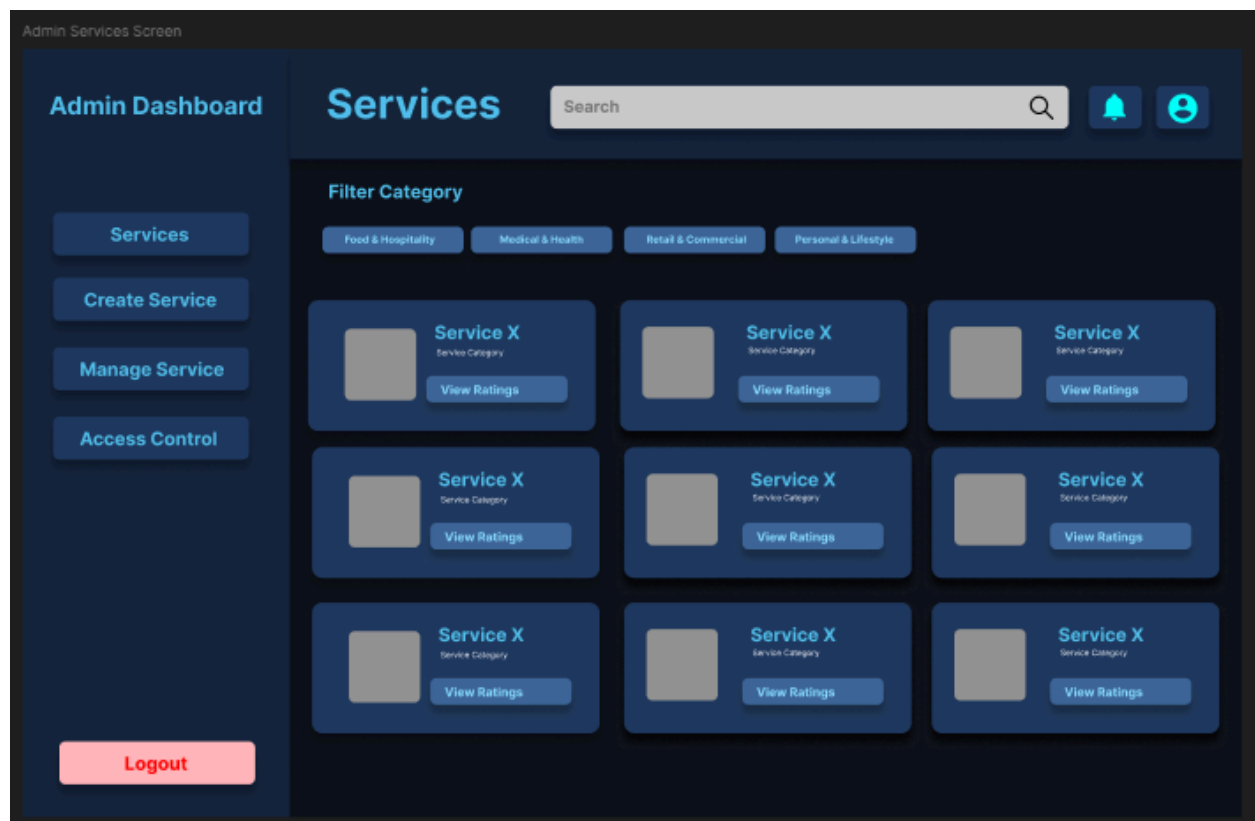
Delete Icon

Layout:

Left Sidebar + Top Navbar + Stacked Notification Cards

Dark-themed UI

Admin Services Page



Header:

Page Title ("Services") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services (Active)
- Create Service
- Manage Service
- Access Control
Logout Button

Content:

Filter Category Buttons (Food & Hospitality, Medical & Health, Retail & Commercial, Personal & Lifestyle)

Services Grid (3 columns desktop)

Each Service Card:

Service Image / Thumbnail

Service Name

Service Category

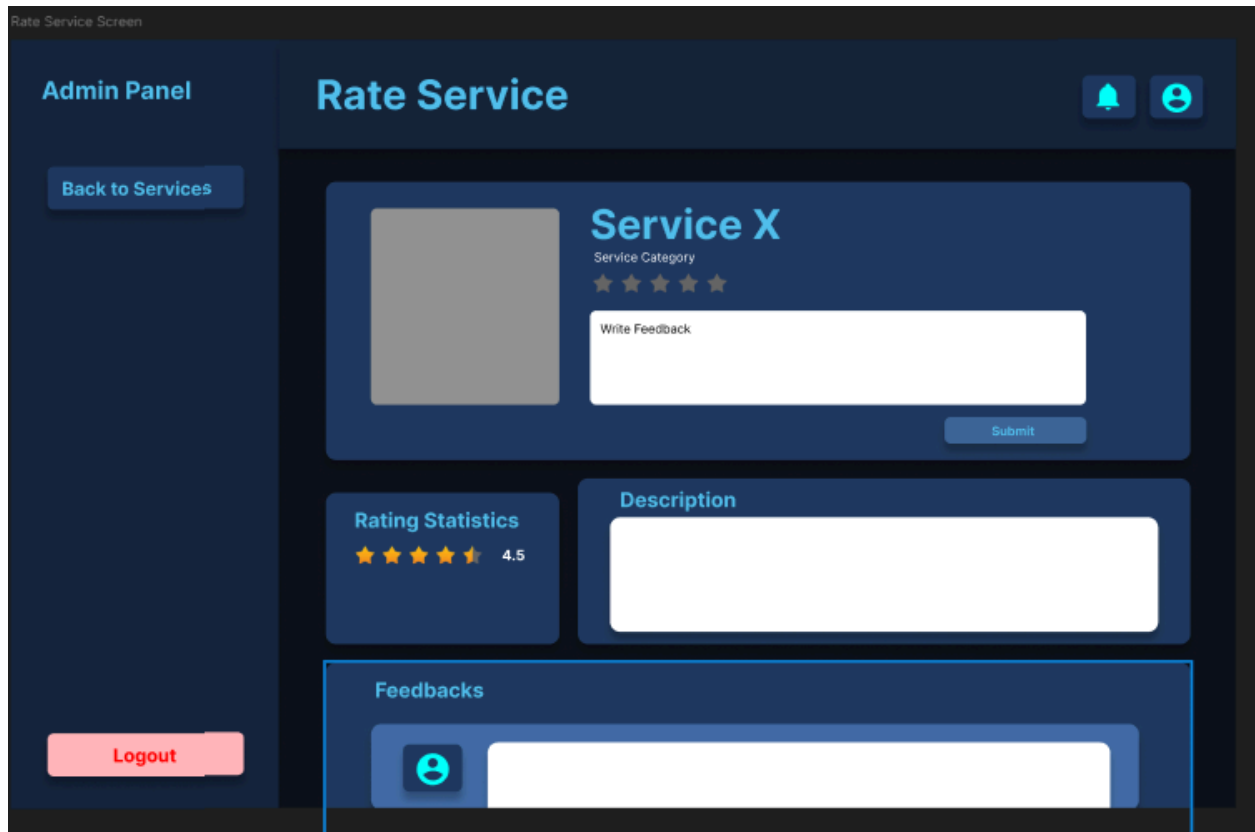
"View Ratings" Button

Layout:

Left Admin Sidebar + Top Navbar + Main Content Grid

Dark-themed UI

Admin View Feedbacks Page



Header:

Page Title ("Ratings") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service
- Access Control
Logout Button

Content:

Service Overview Section:

Service Image / Thumbnail

Service Name
Service Category

Rating Statistics Section:

Star Rating Display
Average Rating Value

Description Section:

Service Description Box

Feedbacks Section (Vertical List):

Feedback Cards

Each Feedback Card:

User Icon
Written Feedback
Delete Button

Layout:

Left Admin Sidebar + Top Navbar + Multi-section Content Layout
Dark-themed UI

Admin Create Service Page

Admin Services Screen

Admin Dashboard

Create Search 🔍 🔔 👤

Services

Create Service

Manage Service

Access Control

Logout

Create Service

Service Name

Category

Service Description

Cancel Create

Header:

Page Title ("Create") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service (Active)
- Manage Service
- Access Control
- Logout Button

Content (Centered Form Card):

Service Image Upload / Placeholder

Form Fields:

Service Name Input

Category Dropdown

Service Description Text Area

Actions:

Cancel Button

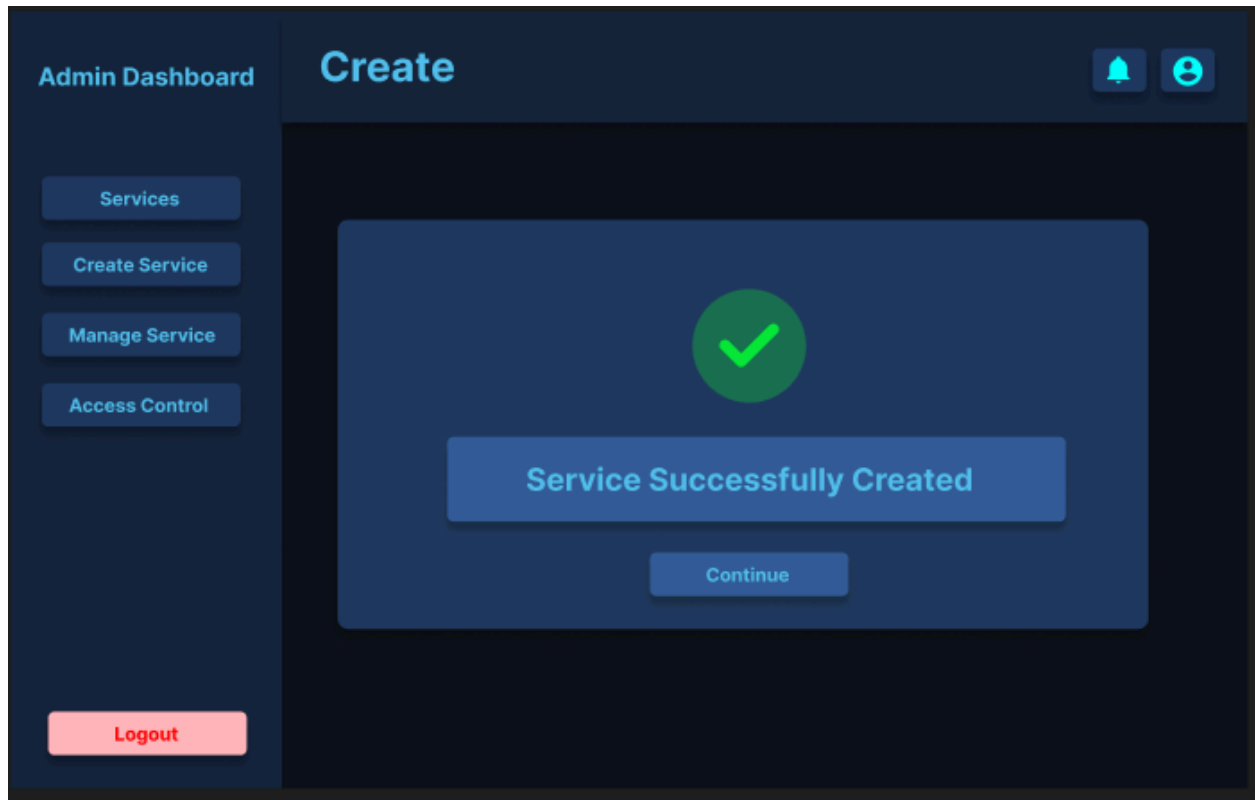
Create Button

Layout:

Left Admin Sidebar + Top Navbar + Centered Form Card

Dark-themed UI

Admin Service Creation Confirmation Page

**Header:**

Page Title ("Create") | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
 - Create Service
 - Manage Service
 - Access Control
- Logout Button

Content (Centered Confirmation Card):

Success Icon (Checkmark)

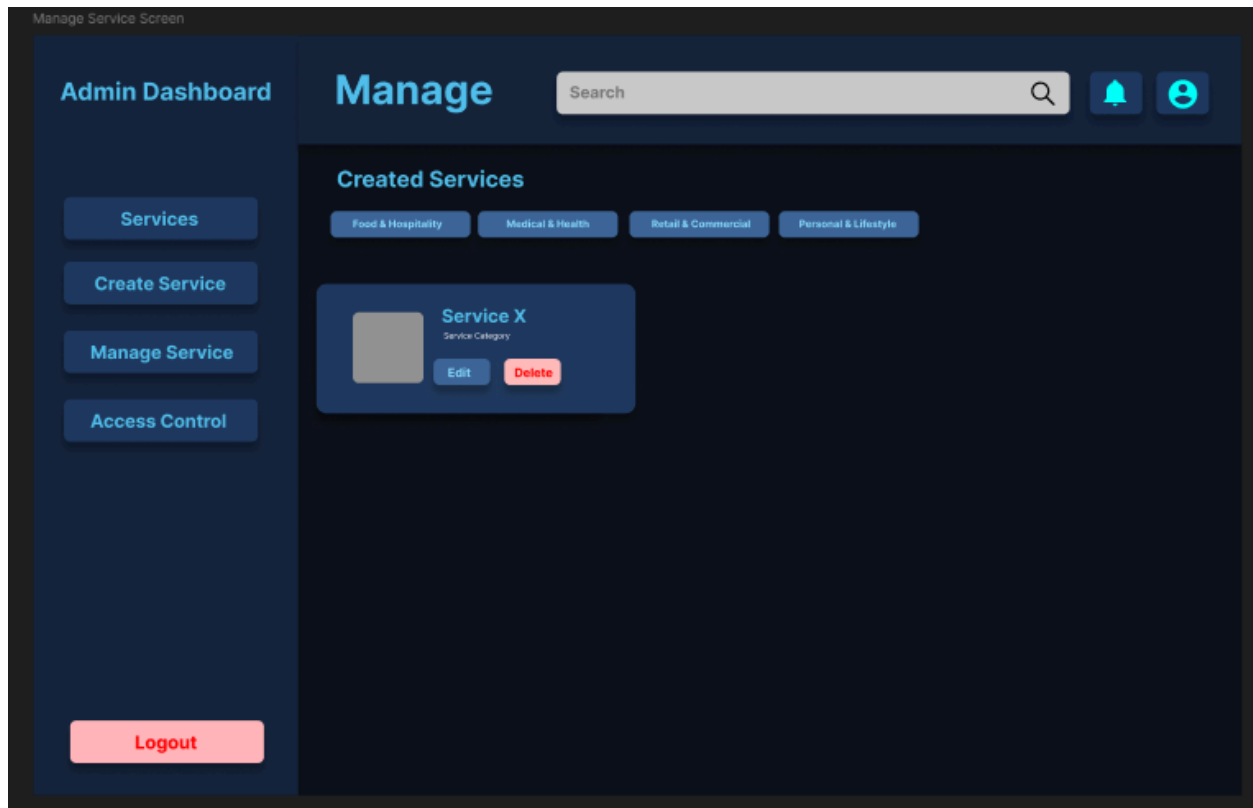
Message: "Service Successfully Created"

"Continue" Button

Layout:

Left Admin Sidebar + Top Navbar + Centered Confirmation Card
Dark-themed UI

Admin Manage Service Page



Header:

Page Title ("Manage") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service (Active)

- Access Control
Logout Button

Content:

Section Title: "Created Services"

Filter Category Buttons (Food & Hospitality, Medical & Health, Retail & Commercial, Personal & Lifestyle)

Services List / Grid

Each Service Card:

Service Image / Thumbnail

Service Name

Service Category

"Edit Service" Button

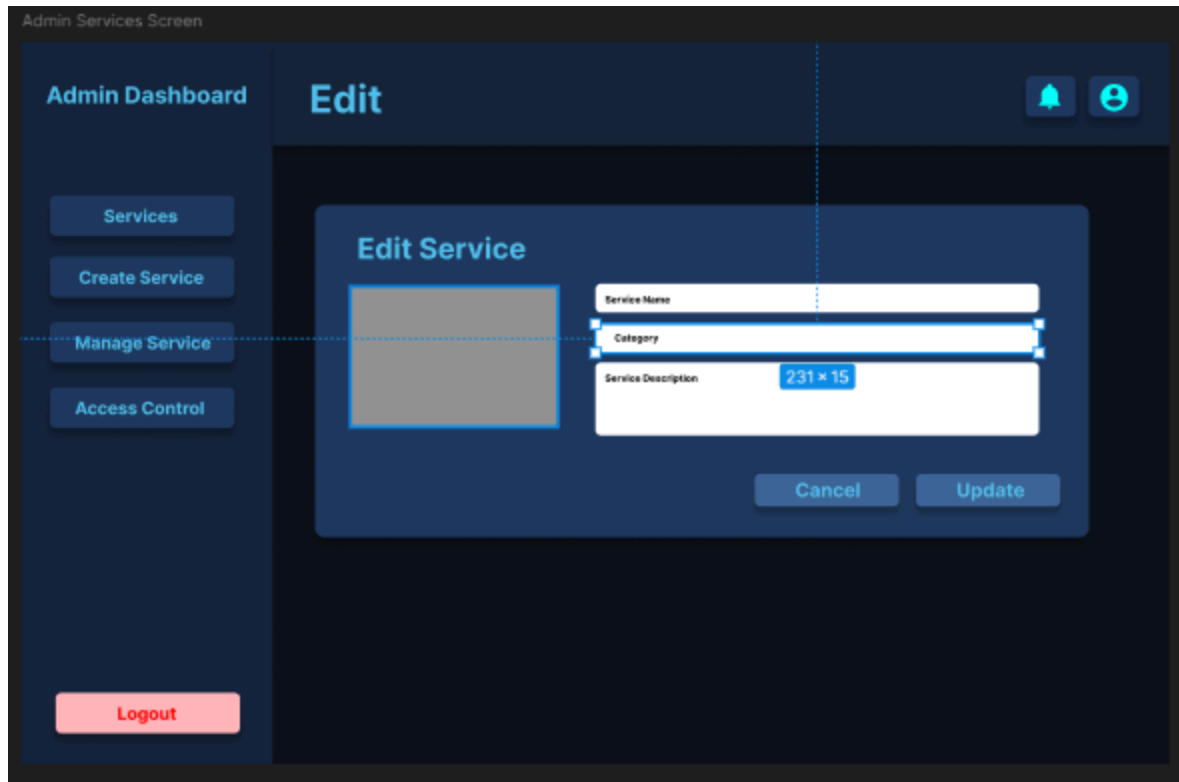
Delete Icon

Layout:

Left Admin Sidebar + Top Navbar + Service Management Grid

Dark-themed UI

Admin Edit Service Page



Header:

Page Title ("Edit") | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service
- Access Control
- Logout Button

Content (Centered Form Card):

Service Image Display / Upload Option

Form Fields:

Service Name Input (Pre-filled)

Category Dropdown (Pre-selected)
Service Description Text Area (Pre-filled)

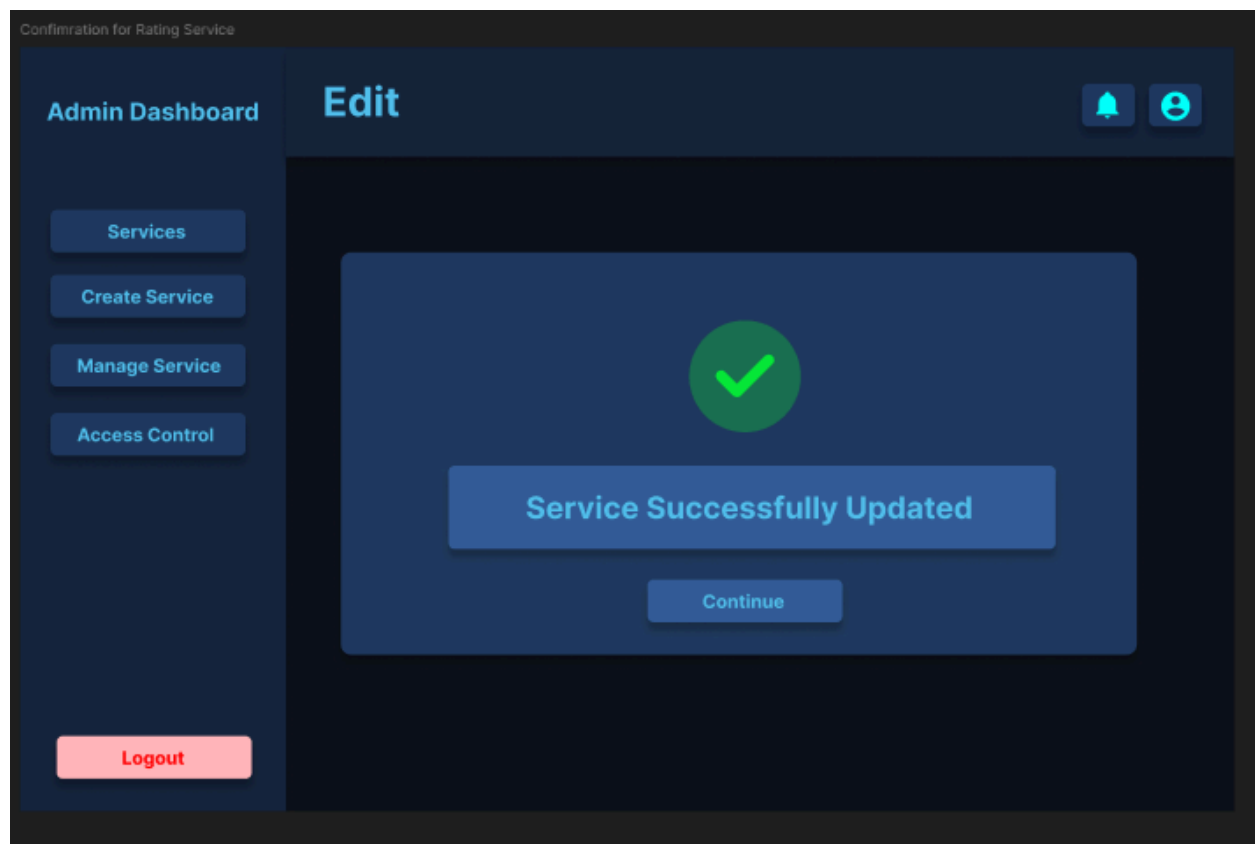
Actions:

Cancel Button
Save Button

Layout:

Left Admin Sidebar + Top Navbar + Centered Edit Form Card
Dark-themed UI

Admin Edit Service Confirmation Page



Header:

Page Title ("Edit") | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service
- Access Control
Logout Button

Content (Centered Confirmation Card):

Success Icon (Checkmark)

Message: "Service Successfully Updated"

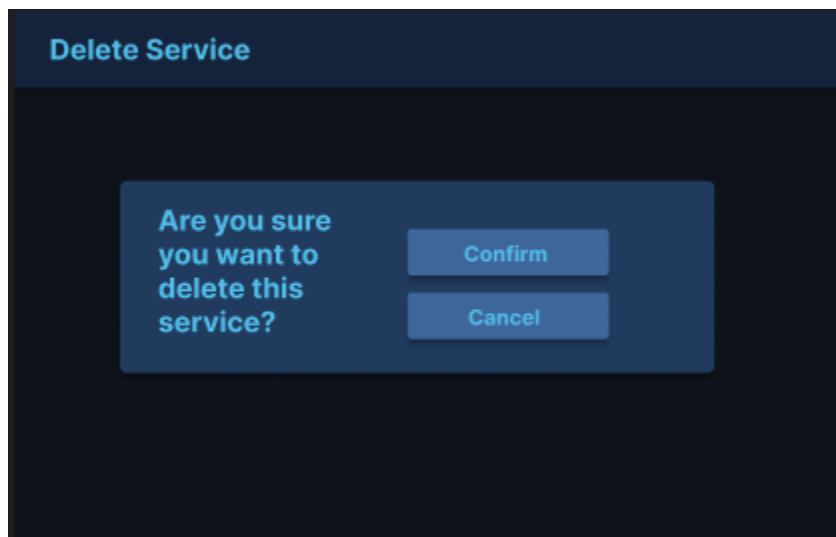
"Continue" Button

Layout:

Left Admin Sidebar + Top Navbar + Centered Confirmation Card

Dark-themed UI

Admin Delete Service Confirmation Page



Header:

Page Title ("Delete Service")

Content (Centered Confirmation Card):

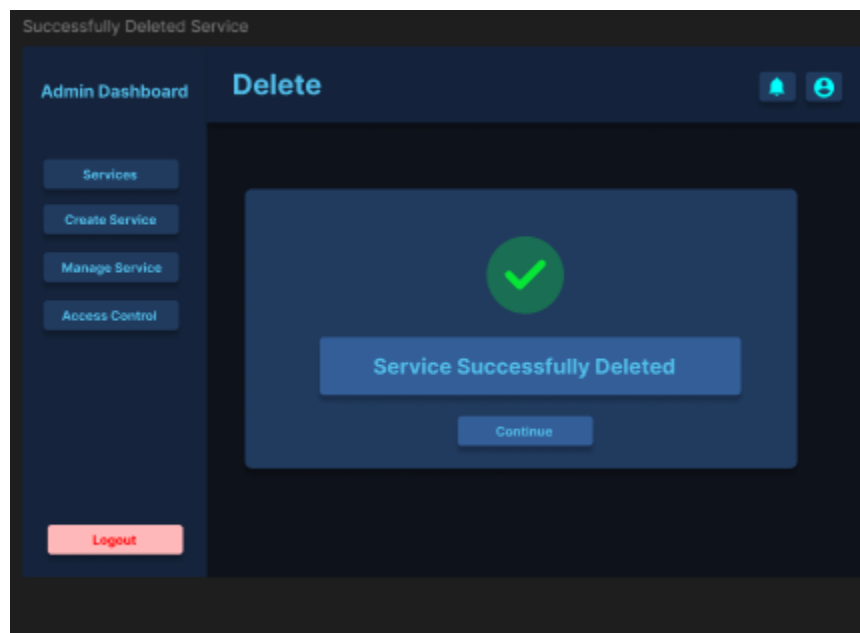
Message: "Are you sure you want to delete this service?"

Confirm Button
Cancel Button

Layout:

Top Header + Centered Confirmation Card
Dark-themed UI

Admin Service Deletion Success Page



Header:

Page Title ("Delete") | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service
- Access Control
- Logout Button

Content (Centered Confirmation Card):

Success Icon (Checkmark)

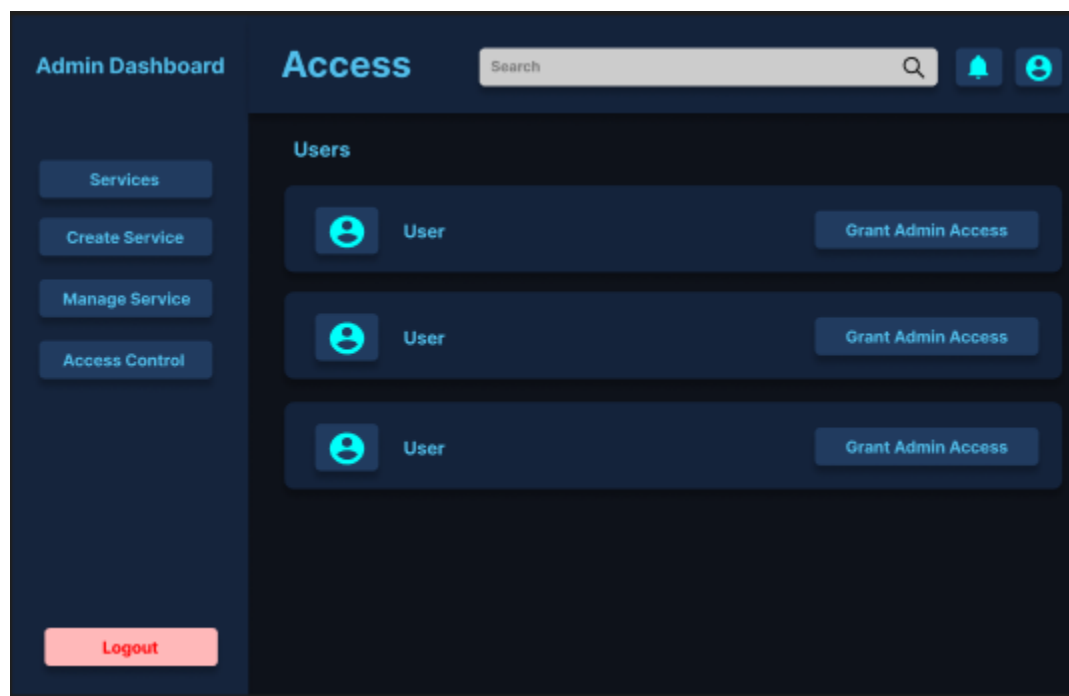
Message: "Service Successfully Deleted"

"Continue" Button

Layout:

Left Admin Sidebar + Top Navbar + Centered Confirmation Card

Dark-themed UI

Admin Access Control Page**Header:**

Page Title ("Access") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service

- Access Control (Active)
Logout Button

Content:

Section Title: "Users"

Users List (Vertical Layout)

Each User Card:

User Icon

Username

"Grant Admin Access" Button

Layout:

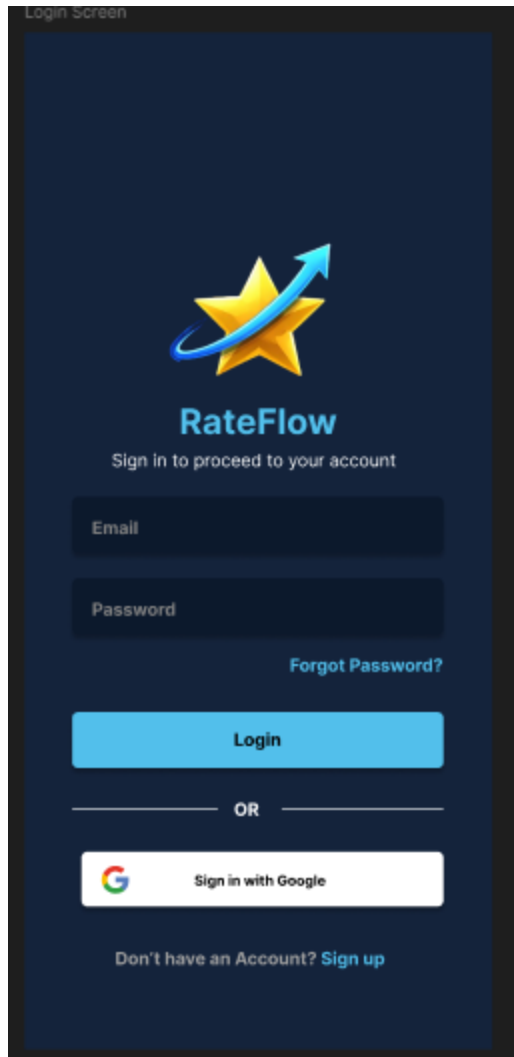
Left Admin Sidebar + Top Navbar + Stacked User Cards

Dark-themed UI

7.2 Mobile Application Wireframes

Note: This should be wireframes from Figma

Mobile Login Page



Header:

App Logo

App Name ("RateFlow")

Subtitle: "Sign in to proceed to your account"

Content (Single Column Layout):

Email Input Field

Password Input Field

"Forgot Password?" Link

Primary Button:

"Login" Button

Divider:

"OR" Separator Line

Secondary Action:

"Don't have an Account? Sign up" Link

"Sign In with Google" Button

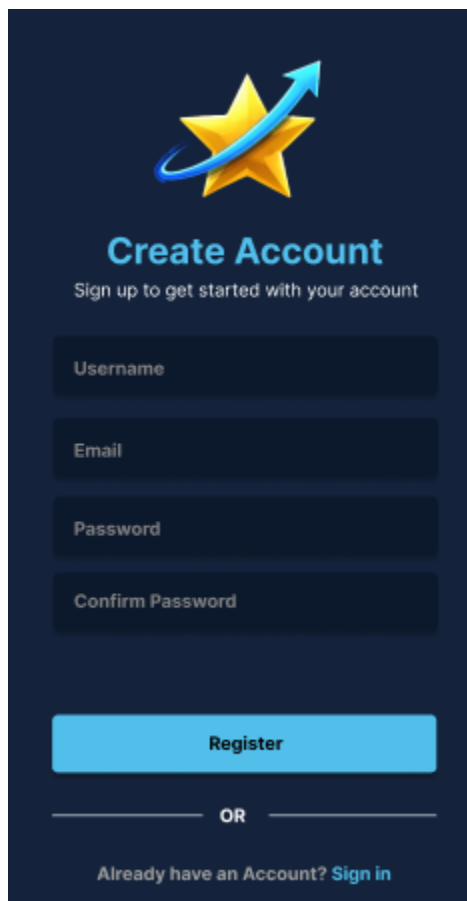
Layout:

Centered Vertical Layout (Single Column)

Full-screen Dark-themed UI

Mobile-optimized spacing and touch-friendly buttons

Mobile Registration Page

A mobile registration page UI mockup with a dark blue background. At the top is a logo featuring a yellow star with a blue arrow curving around it. Below the logo is the title "Create Account" in light blue, followed by the subtitle "Sign up to get started with your account" in white. The form consists of four input fields: "Username", "Email", "Password", and "Confirm Password", each with a light blue placeholder text. Below these fields is a prominent light blue "Register" button. Underneath the button is a horizontal line with the word "OR" in the center. At the bottom, there is a link that says "Already have an Account? Sign in" in light blue.

Header:

App Logo

Title: "Create Account"

Subtitle: "Sign up to get started with your account"

Content (Single Column Layout):

Username Input Field

Email Input Field

Password Input Field

Confirm Password Input Field

Primary Button:

"Register" Button

Divider:

"OR" Separator Line

Secondary Action:

"Already have an Account? Sign in" Link

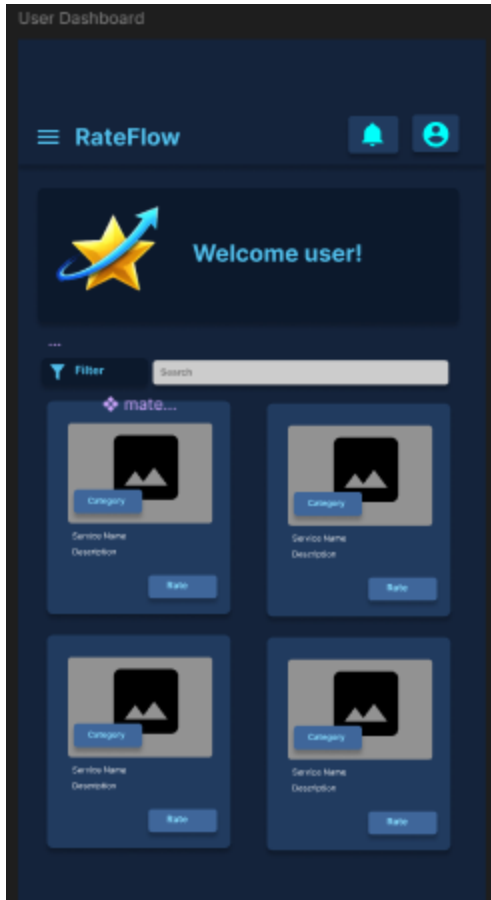
Layout:

Centered Vertical Layout (Single Column)

Full-screen Dark-themed UI

Mobile-optimized spacing and touch-friendly buttons

Mobile User Dashboard Page



Header Section:

Navigation Menu Icon | Application Name ("RateFlow") | Notification Icon | User Profile Avatar

Main Content (Scrollable Layout):

Welcome Banner:

- RateFlow Logo
- Greeting Text: "Welcome user!"

Filter & Search Area:

- Filter Button
- Search Input Field

Service Listing Section (2-Column Mobile Grid):

Each Service Card Contains:

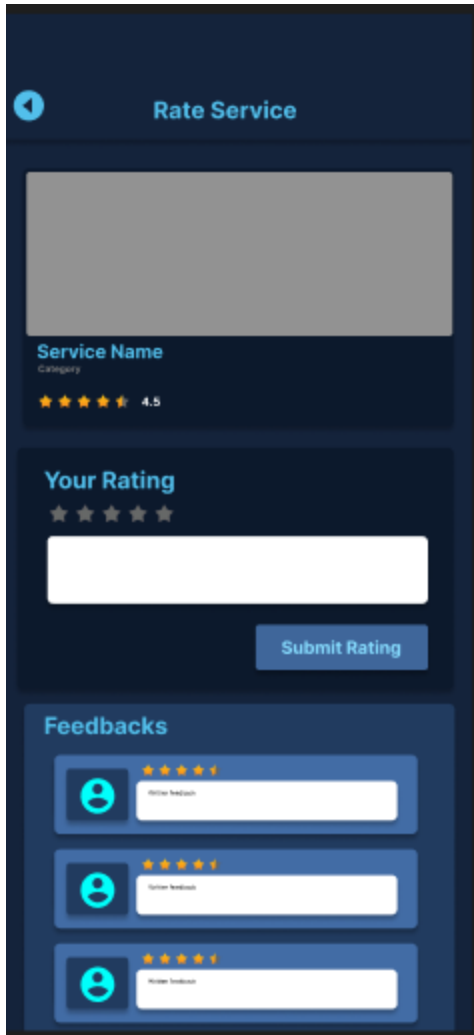
- Service Thumbnail/Image

- Category Label
- Service Name
- Short Description
- “Rate” Button

Overall Layout Design:

- Fixed Top Navigation Bar
- Vertically Scrollable Main Content
- Responsive Two-Column Card Layout
- Dark Blue Modern Theme
- Mobile-Friendly Spacing and Alignment

Mobile Submit Rating Page



Header Section:

Application Title (“Rate Service”) with a Back Navigation Button

Main Content (Scrollable Single Column Layout):

Service Information Section

- Service Image / Thumbnail Preview
- Service Name
- Service Category Label
- Average Rating Display (Star Icons with Numerical Rating)

User Rating Section

- Interactive Star Rating Selection

- Feedback Text Input Area
- “Submit Rating” Action Button

Feedbacks Section

- Vertical List of User Feedback Cards

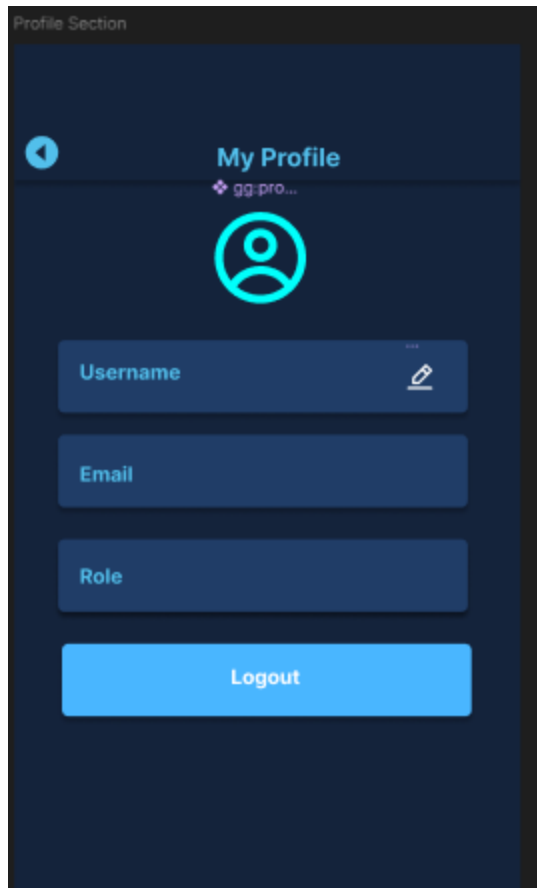
Each Feedback Card Includes:

- User Profile Icon
- Star Rating Display
- User Written Feedback

Overall Layout Design:

- Fixed Top Header with Back Button
 - Fully Scrollable Content Area
 - Single Column Mobile Layout
 - Dark Blue Modern User Interface
 - Mobile-optimized spacing and touch-friendly components
-

Mobile User Profile Page



Header Section:

Page Title ("My Profile") with Back Navigation Button

Main Content (Single Column Layout):

Profile Information Section

- User Profile Avatar/Icon
- Username Display
- User Email Display
- User Role Display
- Edit Profile Button/Icon

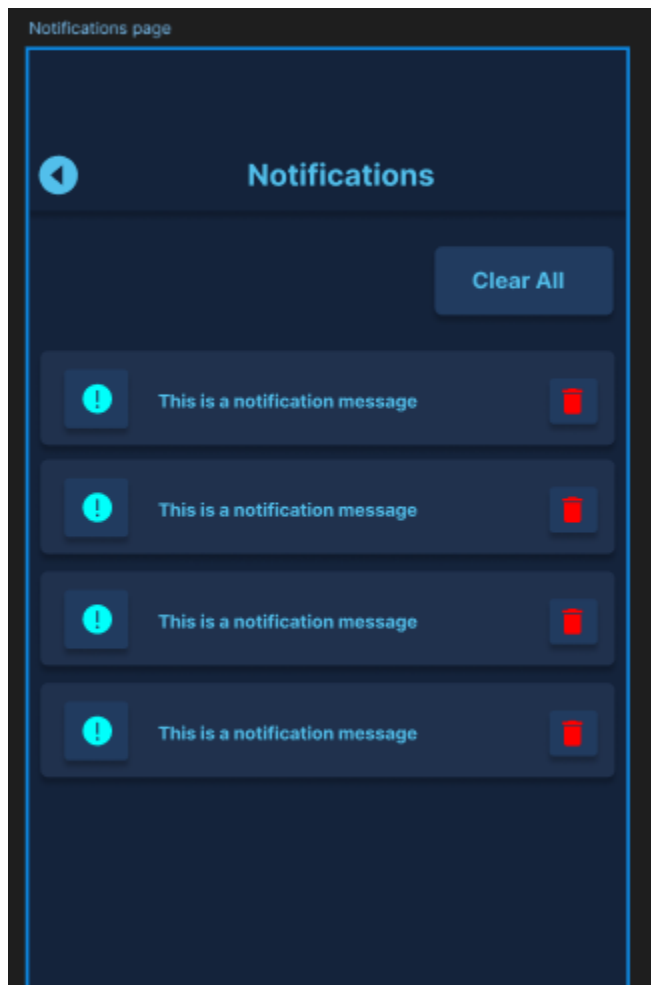
Account Actions

- "Logout" Button

Overall Layout Design:

- Fixed Top Header with Back Button
- Centered Profile Information Card
- Single Column Mobile Layout
- Dark Blue Modern Interface
- Mobile-optimized spacing and touch-friendly controls

Mobile Notifications Page



Header Section:

Page Title ("Notifications") with Back Navigation Button

Main Content (Scrollable Single Column Layout):

Notification Controls

- “Clear All” Button for removing all notifications

Notifications List Section

- Vertical List of Notification Cards

Each Notification Card Includes:

- Notification Status/Icon
- Notification Message Preview
- Delete/Remove Icon Button

Overall Layout Design:

- Fixed Top Header with Back Button
 - Fully Scrollable Content Area
 - Stacked Card-Based Notification Layout
 - Dark Blue Modern User Interface
 - Mobile-optimized spacing with touch-friendly controls
-

Drawer Header

- User Display Name / Account Section

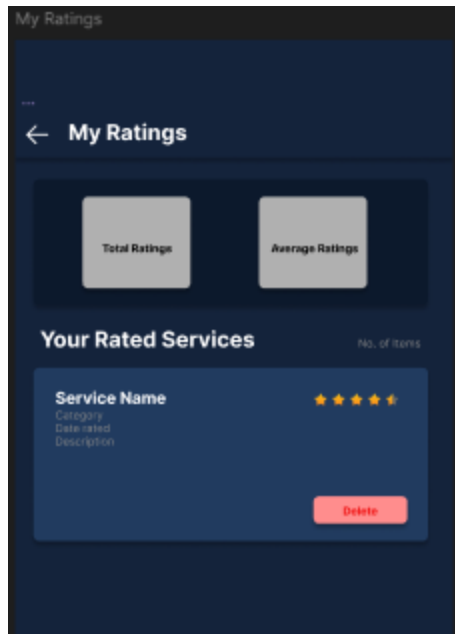
Navigation Menu Items

- Services
- My Ratings

Layout Design

- Slide-in Left Navigation Drawer
 - Overlay displayed above the main dashboard content
 - Dark Blue Modern User Interface
 - Large touch-friendly navigation buttons for mobile interaction
-

Mobile My Ratings Page



Header Section:

Page Title (“My Ratings”) with Back Navigation Button

Main Content (Scrollable Single Column Layout):

Ratings Overview Section

- Summary Cards displaying:
 - Total Ratings
 - Average Ratings

Rated Services Section

- Vertical List of Rated Service Cards

Each Rating Card Includes:

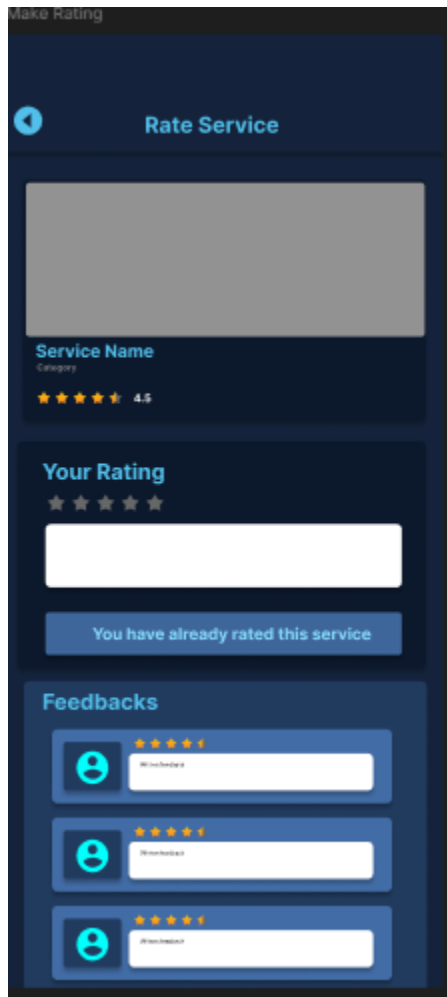
- Service Name
- Service Category
- Date Rated Information
- Star Rating Display
- “Delete” Rating Button

Overall Layout Design:

- Fixed Top Header with Back Button

- Fully Scrollable Content Area
- Card-Based Rating Layout
- Dark Blue Modern User Interface
- Mobile-optimized spacing and touch-friendly controls

Mobile View Rating Page



Header Section:

Page Title ("Rate Service") with Back Navigation Button

Main Content (Scrollable Single Column Layout):

Service Details Section

- Service Image / Thumbnail

- Service Name
- Service Category Label
- Average Rating Display (Star Icons with Numerical Rating)

User Rating Section

- User Star Rating Display
- Written Feedback Display Area
- Status Message indicating the service has already been rated

Feedbacks Section

- Vertical List of User Feedback Cards

Each Feedback Card Includes:

- User Profile Icon
- Star Rating Display
- Written Feedback Content

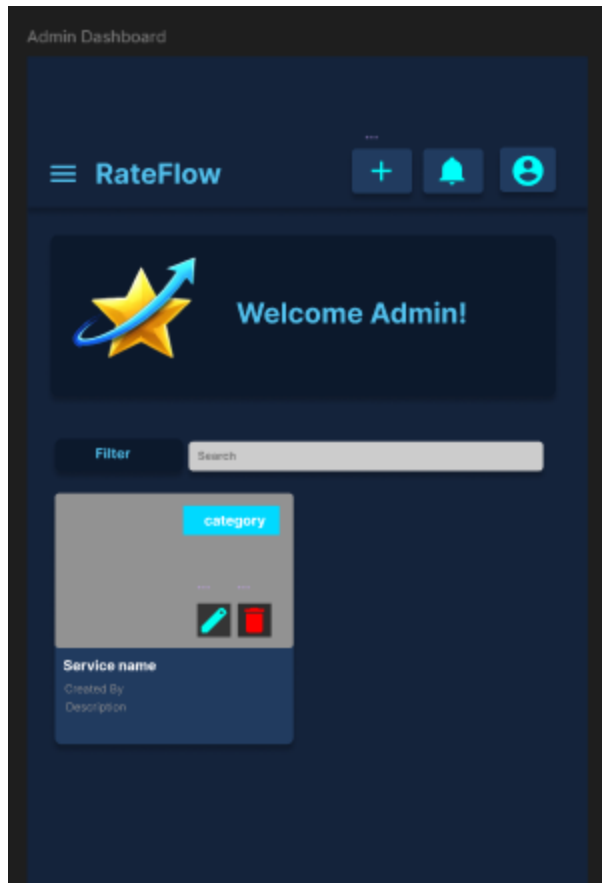
Available Actions

- Edit Rating Icon/Button
- Delete Rating Icon/Button

Overall Layout Design:

- Fixed Top Header with Back Navigation Button
- Fully Scrollable Content Area
- Single Column Mobile Layout
- Dark Blue Modern User Interface
- Mobile-optimized spacing with touch-friendly controls

Mobile Admin Dashboard Page



Header Section:

Navigation Menu Icon | Application Name ("RateFlow") | Notification Icon | Admin Profile Icon | Create Service Icon "+"

Main Content (Scrollable Single Column Layout):

Welcome Banner

- RateFlow Logo
- Greeting Message: "Welcome Admin!"

Filter & Search Section

- Filter Button
- Search Input Field

Service Management Section (Card Grid Layout)

- Responsive Service Card Display

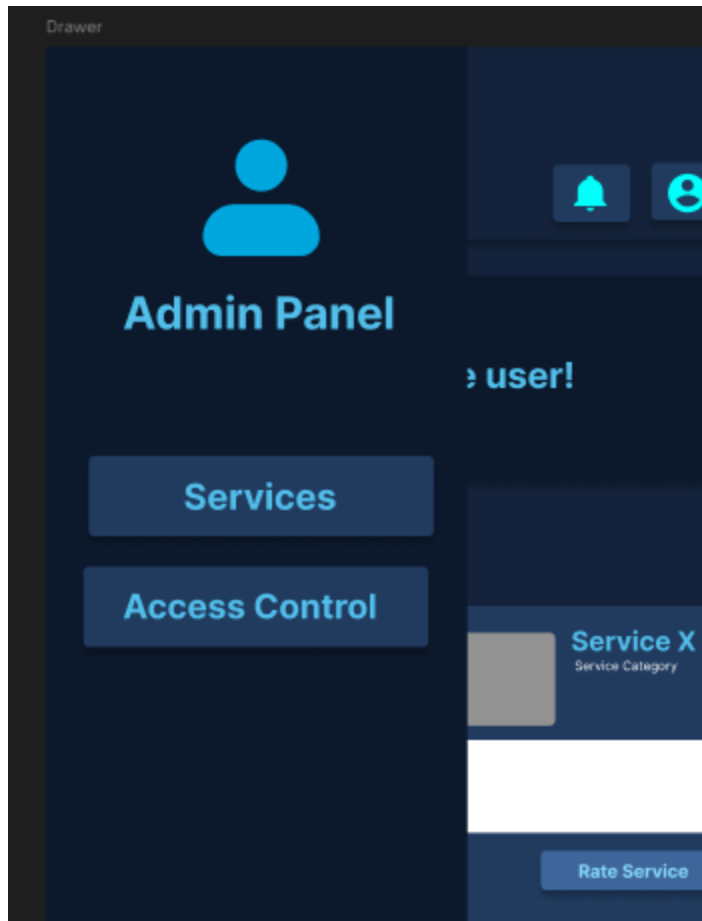
Each Service Card Includes:

- Service Thumbnail/Image
- Service Category Label
- Service Name
- Creator Information
- Short Service Description
- Edit Service Button/Icon
- Delete Service Button/Icon

Overall Layout Design:

- Fixed Top Navigation Bar
- Fully Scrollable Content Area
- Responsive Card-Based Service Layout
- Dark Blue Modern User Interface
- Mobile-optimized spacing with touch-friendly controls

Mobile Admin Drawer (Side Menu)



Drawer Header

- Admin Profile Icon
- Title: "Admin Panel"

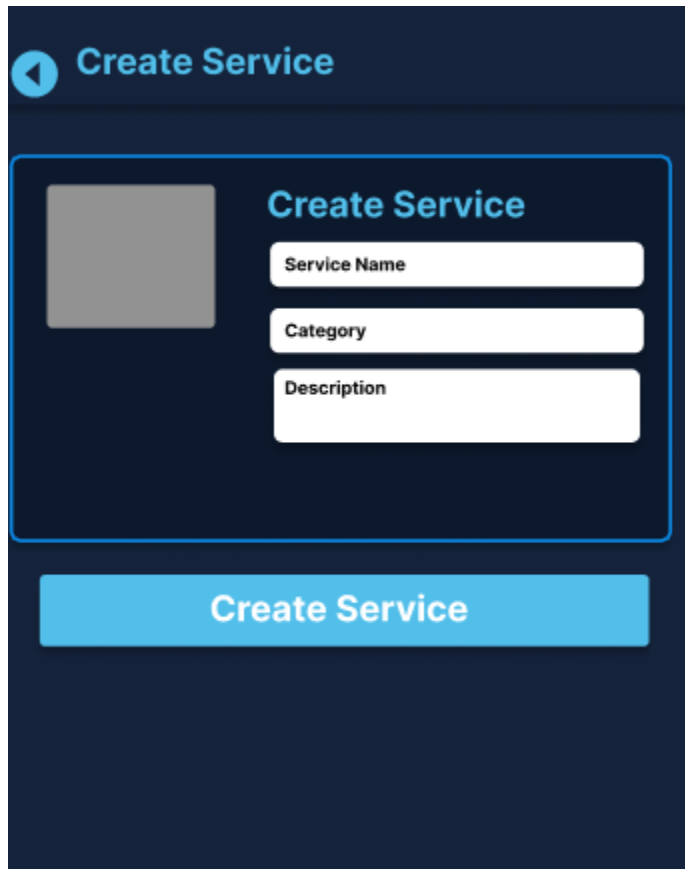
Navigation Menu Items

- Services
- Access Control

Layout Design

- Slide-in Left Navigation Drawer
- Overlay displayed above the main dashboard content
- Dark Blue Modern User Interface
- Large touch-friendly navigation buttons for mobile interaction

Mobile Admin – Create Service Page



The image shows a mobile application interface for creating a service. At the top, there is a dark blue header bar with a back navigation button (a white circle with a left-pointing arrow) and the title "Create Service" in white. Below the header, the main content area has a dark blue background. It features a rounded rectangle containing a service image placeholder (a gray square) on the left and a form on the right. The form is titled "Create Service" in white and includes three input fields: "Service Name", "Category", and "Description". Below this rounded rectangle is a large, light blue button with the text "Create Service" in white.

Header Section:

Page Title ("Create Service") with Back Navigation Button

Main Content (Single Column Layout):

Create Service Form Section

- Service Image Upload / Image Placeholder
- Service Name Input Field
- Category Selection Dropdown
- Description Text Area

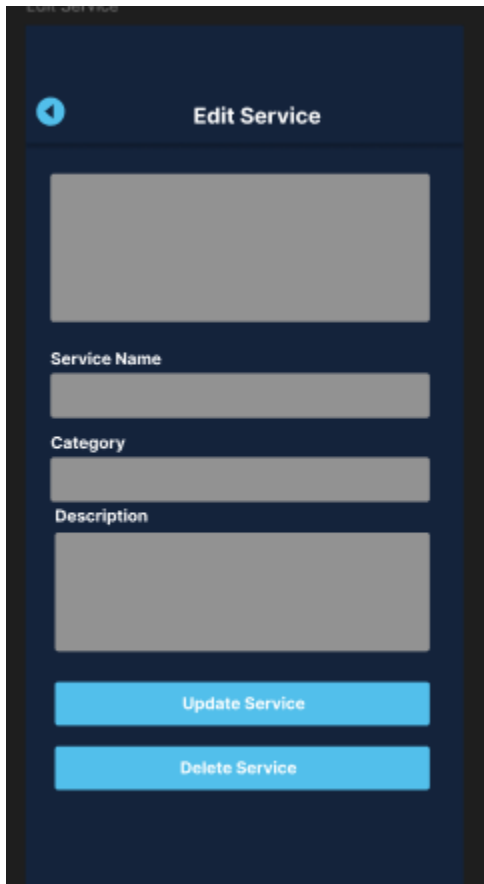
Action Button

- "Create Service" Button

Overall Layout Design:

- Fixed Top Header with Back Navigation Button
 - Centered Form Card Layout
 - Single Column Mobile Interface
 - Dark Blue Modern User Interface
 - Mobile-optimized spacing with touch-friendly form controls
-

Mobile Admin – Edit Service Page



The image shows a mobile application interface for editing a service. It features a dark blue header with a back navigation button (a circle with a left-pointing arrow) and the title "Edit Service". Below the header is a large, empty gray rectangular area. Underneath this is a form with three input fields: "Service Name", "Category", and "Description". The "Description" field is a larger text area. At the bottom of the form are two blue buttons: "Update Service" and "Delete Service".

Header Section:

Page Title ("Edit Service") with Back Navigation Button

Main Content (Single Column Layout):

Edit Service Form Section

- Service Image Preview / Thumbnail
- Service Name Input Field
- Service Category Input Field
- Service Description Text Area

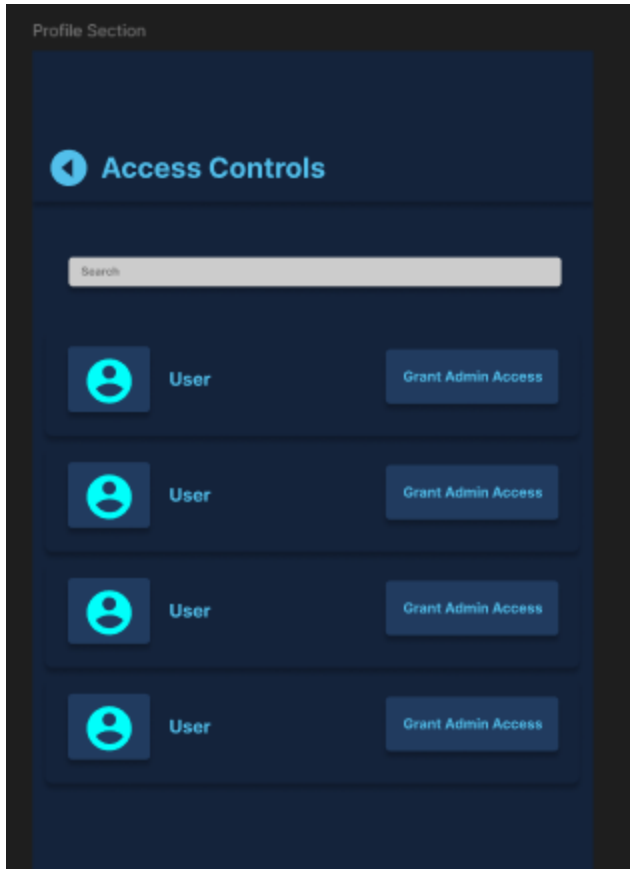
Action Buttons

- “Update Service” Button
- “Delete Service” Button

Overall Layout Design:

- Fixed Top Header with Back Navigation Button
- Single Column Form Layout
- Dark Blue Modern User Interface
- Fully Scrollable Content Area
- Mobile-optimized spacing with touch-friendly controls

Mobile Admin – Access Control Page



Header Section:

Page Title ("Access Controls") with Back Navigation Button

Main Content (Scrollable Single Column Layout):

Search Section

- Search Input Field for filtering users

User Management Section

- Vertical List of User Cards

Each User Card Includes:

- User Profile Icon
- Username Display
- "Grant Admin Access" Button

Overall Layout Design:

- Fixed Top Header with Back Navigation Button
 - Fully Scrollable Content Area
 - Stacked Card-Based User Management Layout
 - Dark Blue Modern User Interface
 - Mobile-optimized spacing with touch-friendly controls
-

Design System

Color Palette

- **Primary:** #52BFEB — Main buttons, highlights, active states, icons
 - **Secondary:** #223A5E — Cards, elevated containers, navigation buttons
 - **Background:** #121A2B — Main dark application background
 - **Surface:** #0D223F — Panels, banners, drawer surfaces, section containers
 - **Success:** #22C55E — Success messages, confirmations, completed actions
 - **Error:** #EF4444 — Delete actions, warnings, destructive operations
 - **Text Primary:** #E6F0FF — Main readable text
 - **Text Secondary:** #7A8499 — Secondary labels, placeholders, descriptions
 - **Border / Divider:** #2A3448 — Section dividers, outlines, subtle borders
-

Typography

- **Font Family:** Inter, sans-serif
- **Headings:** Bold with responsive scaling
 - H1 → Page Titles
 - H2 → Section Headers
 - H3 → Card Titles
 - H4 → Labels & Subsections
- **Body Text:** Regular / Medium weight
- **Buttons:** Semi-bold, high readability
- **Navigation Labels:** Medium weight with mobile-friendly sizing

Spacing System

- Uses an **8px grid system**
- Standard spacing scale:
 - 8px
 - 16px
 - 24px
 - 32px
 - 40px
 - 48px

Layout Consistency

- Uniform card padding
 - Balanced vertical spacing between sections
 - Touch-friendly margins and button spacing
 - Consistent alignment across mobile screens
-

UI Components

Buttons

- Primary Button
- Secondary Button
- Danger/Delete Button
- Disabled State Button

Button Style

- Rounded corners
- Elevated dark-theme appearance
- Touch-friendly sizing
- Subtle shadow effects

Input Fields

- Rounded dark-themed input containers
 - Focus glow/highlight effect
 - Search bars, dropdowns, and text areas follow consistent styling
 - Mobile-friendly text sizing and padding
-

Cards

- Rounded corner layout
 - Soft elevation shadow
 - Dark surface styling
 - Used for:
 - Service Cards
 - Feedback Cards
 - Notification Cards
 - User Cards
 - Summary Cards
-

Navigation Drawer

- Slide-in left drawer layout
 - Overlay above main content
 - Dark modern surface styling
 - Large touch-friendly navigation buttons
 - Back arrow collapse button for drawer closing
-

Modals & Dialogs

- Centered overlay layout
- Dimmed dark background
- Rounded card appearance

- Focused action buttons
-

Icons

- Consistent rounded icon style
 - Cyan-blue accent color
 - Minimal flat icon design
 - Optimized for mobile readability
-

Layout Structure

Navigation

- Fixed top header
- Back navigation button on inner pages
- Notification and profile quick actions

Content

- Fully scrollable main content
 - Single-column mobile-first layouts
 - Card-based section organization
-

Responsive Design

Mobile-First Approach

Breakpoints

- **640px** → Small Tablets
- **768px** → Tablets

- **1024px** → Desktop Screens
-

Grid Behavior

- **Mobile:** 1-column layout
 - **Tablet:** 2-column card grid
 - **Desktop:** 3-column grid layout
-

User Experience Principles

- Mobile-optimized spacing
- Touch-friendly controls
- Clear visual hierarchy
- Consistent dark-theme appearance
- Minimalist modern interface
- Smooth navigation flow
- Readable typography and accessible contrast

8.0 PLAN

8.1 Project Timeline

Phase 1: Planning & Design (Week 1-2)

Week 1: Requirements & Architecture

Day 1-2: Project setup and documentation

Day 3-4: Complete FRS and NFR

Day 5-7: System architecture design

Week 2: Detailed Design

Day 1-2: API specification

Day 3-4: Database design

Day 5-6: UI/UX wireframes

Day 7: Implementation plan finalization

Phase 2: Backend Development (Week 3-4)

Week 3: Foundation

Day 1: Spring Boot setup with dependencies

Day 2: Database configuration and entities

Day 3: JWT authentication implementation

Day 4: User management endpoints

Day 5: Product CRUD operations

Week 4: Core Features

Day 1: Cart functionality

Day 2: Order management

Day 3: Search and filtering

Day 4: Error handling and validation

Day 5: API documentation and testing

Phase 3: Web Application (Week 5-6)

Week 5: Frontend Foundation

Day 1: React setup with TypeScript

Day 2: Authentication pages (login, register)

Day 3: Product listing page

Day 4: Product detail page

Day 5: Shopping cart implementation

Week 6: Complete Web Features

Day 1: Checkout flow

Day 2: Order history and confirmation

Day 3: Admin dashboard

Day 4: Responsive design polish

Day 5: API integration and testing

Phase 4: Mobile Application (Week 7-8)

Week 7: Android Foundation

Day 1: Android Studio setup and project structure

Day 2: Authentication screens

Day 3: Product browsing

Day 4: Shopping cart

Day 5: API service layer

Week 8: Complete Mobile App

Day 1: Checkout flow

Day 2: Order management

Day 3: UI polish and animations

Day 4: Testing on emulator/device

Day 5: APK generation and documentation

Phase 5: Integration & Deployment (Week 9-10)

Week 9: Integration Testing

Day 1: End-to-end testing across platforms

Day 2: Bug fixes and optimization

Day 3: Security review

Day 4: Performance testing

Day 5: Documentation updates

Week 10: Deployment

Day 1: Backend deployment (Railway/Heroku)

Day 2: Web app deployment (Vercel/Netlify)

Day 3: Mobile APK distribution

Day 4: Final testing

Day 5: Project submission

Milestones:

- **M1 (End Week 2):** All design documents complete
- **M2 (End Week 4):** Backend API fully functional
- **M3 (End Week 6):** Web application complete
- **M4 (End Week 8):** Mobile application complete
- **M5 (End Week 10):** Full system deployed and integrated

Critical Path:

1. Authentication system (Week 3)
2. Product catalog API (Week 3-4)
3. Shopping cart functionality (Week 4)
4. Checkout process (Week 6)
5. Cross-platform testing (Week 9)

Risk Mitigation:

- Start with simplest working version of each feature
- Test integration points early and often
- Keep backup of working versions

- Focus on core functionality before enhancements